8
Functions

The notion of *function* is one of the most fundamental in mathematics. It is probably familiar to the reader in the context of calculus, and it is here that the concept was first clarified and the difference between a function and a formula properly understood in the early years of the nineteenth century. Today, the language of functions is used throughout mathematics.

In this chapter the function concept is introduced. We discuss various ways of defining a function and consider the graph of a function.

8.1 Functions and formulae

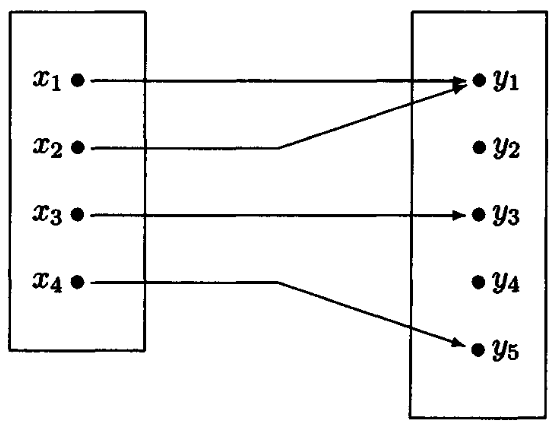
Definition 8.1.1 *Suppose that X and Y are sets. A* $\boxed{\text{function}}$, map *or* mapping *from the set X to the set Y is the assignment of a unique element of Y to each element of X. If f is a function from X to Y we write* $\boxed{f\colon X \to Y}$ *and denote the element of Y assigned to an element $x \in X$ by* $\boxed{f(x)}$ *writing* $\boxed{x \mapsto f(x)}$. *The element $f(x) \in Y$ is called the* $\boxed{\text{value}}$ *of f at $x \in X$ or the* $\boxed{\text{image}}$ *of x under f. The set X is called the* $\boxed{\text{domain}}$ *of the function f and the set Y is called the* $\boxed{\text{codomain}}$.

Example 8.1.2 The most basic way of describing a function is by listing the values. For example, given $X = \{x_1, x_2, x_3, x_4\}$ and $Y = \{y_1, y_2, y_3, y_4, y_5\}$, the following table determines a function $f\colon X \to Y$.

x	x_1	x_2	x_3	x_4
$f(x)$	y_1	y_1	y_3	y_5

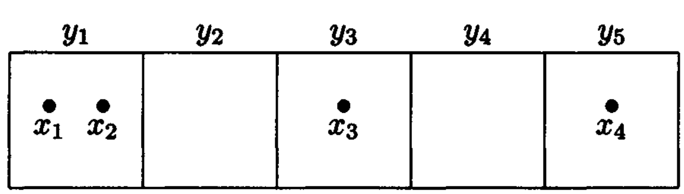
We determine the value of the function at an element of X by locating that element in the first row and reading down to the second row. Notice that each element of the domain X occurs precisely once in the first row so that this is a well-defined procedure. On the other hand elements of the codomain Y may occur once in the second row, but they may also occur more than once or not at all: an element of the codomain may be the value of the function at several elements of the domain or may not be a value at all.

We can picture this function as follows.



Here we find the value of f at an element of the domain X by following the arrow which begins at that element. Each element of the domain lies at the beginning of just one arrow so this is a well-defined procedure. However, elements in the codomain may lie at the end of one arrow, several arrows or no arrow at all.

Yet another way of describing a function is as follows. Think of the elements of the codomain set as boxes. The function describes how to place the objects of the domain set in these boxes. Thus the function under consideration would be pictured as follows.



Again notice that each element of the domain set occurs precisely once in the picture. However, any given box, corresponding to an element of the codomain, may contain one object, several objects or none at all.

The reader may wonder at this variety of ways of thinking about a function. It is important to realize that mathematicians develop a variety of ways of thinking about the rather abstract concepts they deal with. Any given way may suit one person better than another and each leads to different insights: it can be useful to be aware of several.

Example 8.1.3 Suppose that $X = \{a, b, c\}$ and $Y = \{d, e\}$. Then there are precisely eight functions from the set X to the set Y given as follows.

x	$f_1(x)$	$f_2(x)$	$f_3(x)$	$f_4(x)$	$f_5(x)$	$f_6(x)$	$f_7(x)$	$f_8(x)$
a	d	d	d	d	e	e	e	e
b	d	d	e	e	d	d	e	e
c	d	e	d	e	d	e	d	e

Here the functions f_1 and f_8 are examples of *constant functions.* Given sets X and Y and any element $y_0 \in Y$ there is a *constant function* $c_{y_0}: X \to Y$ given by $c_{y_0}(x) = y_0$ for all $x \in X$.

Of course a function can have the same set as domain and codomain as in the next example.

Example 8.1.4 Suppose that $Z = \{a, b\}$. Then there are precisely four functions from Z to Z as follows.

x	$g_1(x)$	$g_2(x)$	$g_3(x)$	$g_4(x)$
a	a	a	b	b
b	a	b	a	b

In this case the functions g_1 and g_4 are constant functions. The function g_2 is an example of an *identity function.* Given a set X, the *identity function* on X, $I_X: X \to X$, is given by $I_X(x) = x$ for all $x \in X$.

Describing a function by listing the values is only practical when the domain set is small and impossible if the domain set is infinite. The most common way of describing a function f is by giving a *formula* for f which provides a procedure for finding the value of the function at each element of the domain. When defining a function using a formula it is important to be clear about which sets are the domain and the codomain of the function.

Example 8.1.5 Recall that $\mathbb{R}^{\geqslant}$ denotes the subset of $\mathbb{R}$, the set of real numbers, given by $\{x \in \mathbb{R} \mid x \geqslant 0\}$. Consider the following four functions:

(i) $f_1: \mathbb{R} \to \mathbb{R}$ given by $f_1(x) = x^2$;
(ii) $f_2: \mathbb{R}^{\geqslant} \to \mathbb{R}$ given by $f_2(x) = x^2$;
(iii) $f_3: \mathbb{R} \to \mathbb{R}^{\geqslant}$ given by $f_3(x) = x^2$;
(iv) $f_4: \mathbb{R}^{\geqslant} \to \mathbb{R}^{\geqslant}$ given by $f_4(x) = x^2$.

Notice that, although the *formula* $f_i(x) = x^2$ is the same for each of these four functions, they are considered to be four distinct functions since the domain and codomain are part of the definition of the function. We will see later that these four functions have different properties even though they are given by the same formula.

We need to take care that the formula makes sense for each element of the domain.

Example 8.1.6 The formula $f_1(x) = \dfrac{x^2 + x - 2}{x - 1}$ does not define a function $\mathbb{R} \to \mathbb{R}$ since it gives no value for $x = 1$. When working with the set of real numbers it is quite common not to specify the domain and codomain of a function given by a formula. The convention is that, if these are not specified, then we take as the domain the subset of $\mathbb{R}$ consisting of the numbers for which the formula makes sense and we take $\mathbb{R}$ as the codomain. Using this convention the above formula for f_1 defines a function $f_1: \mathbb{R} - \{1\} \to \mathbb{R}$.

If we really want a function with domain $\mathbb{R}$ then in an example like this there are two basic techniques for extending f_1.

Rewriting the formula: We can rewrite the formula in such a way that it makes sense for all real numbers x using

$$(x^2 + x - 2)/(x - 1) = (x - 1)(x + 2)/(x - 1) = x + 2$$

for $x \neq 1$. Then $f_2(x) = x + 2$ does define a function on $\mathbb{R}$, i.e. with domain $\mathbb{R}$, extending the function f_1.

Explicit definition: Alternatively we can explicitly specify the value of the function at the element 1 where the formula for f_1 does not work. Thus

$$f_3(x) = \begin{cases} \dfrac{x^2 + x - 2}{x - 1} & \text{if } x \neq 1, \\ 3 & \text{if } x = 1, \end{cases}$$

defines a function $f_3: \mathbb{R} \to \mathbb{R}$.

Notice that we may specify the values at individual points any way we like. It doesn't have to relate in a sensible way to the values elsewhere. In defining $f_3(1)$ we selected that value given by the formula $x + 2$ so that f_2 and f_3 are in fact the same function. However, we could extend the function f_1 to the whole of $\mathbb{R}$ in a different way. For example

$$f_4(x) = \begin{cases} \dfrac{x^2 + x - 2}{x - 1} & \text{if } x \neq 1, \\ 42 & \text{if } x = 1, \end{cases}$$

is another way of extending the function f_1 to the whole of $\mathbb{R}$.

Example 8.1.7 In some cases, for example $x \mapsto 1/x$, there appears to be no way of rewriting the formula to include in the domain of definition points where it gives no value (in this case $x = 0$). But in these cases we can still use the second method. For example

$$g(x) = \begin{cases} 1/x & \text{if } x \neq 0, \\ 73 & \text{if } x = 0, \end{cases}$$

is a perfectly respectable function $g: \mathbb{R} \to \mathbb{R}$.

We can use different formulae for different parts of the domain as in the following example.

Example 8.1.8 The *modulus function* $x \mapsto |x|$ is defined on $\mathbb{R}$ by

$$|x| = \begin{cases} x & \text{if } x \geqslant 0, \\ -x & \text{if } x \leqslant 0. \end{cases}$$

Notice here that the value of the function at 0 is given twice, since $0 \geqslant 0$ and $0 \leqslant 0$. This isn't a problem since the two formulae ($x \mapsto x$ and $x \mapsto -x$) give the same value (namely 0) for $x = 0$. But when specifying a function in this way we do need to be careful to check that it is *well-defined* (i.e. there is a uniquely specified value) at each point.

Definition 8.1.9 *Two functions $f: X \to Y$ and $g: X \to Y$ are* $\boxed{\text{equal}}$, *written* $\boxed{f = g}$, *when they have the same value at each point of the domain X, i.e. $f(x) = g(x)$ for all $x \in X$. Notice that it is implicit in this definition that two equal functions have the same domain and the same codomain.*

Example 8.1.10 The functions f_2 and f_3 of Example 8.1.6 are equal even though they are defined in different ways. Thus it is not the process leading to the values which determines the function but what the values are. Of course, in defining f_3 the value of $f_3(1)$ was selected precisely so that they would be equal.

We have seen in Example 8.1.6 how to extend the domain of a function. We can also make the domain smaller.

Definition 8.1.11 *Suppose that $f: X \to Y$ is a function and A is a subset of X, i.e. $A \subseteq X$. Then we can define a function $g: A \to Y$ by $g(a) = f(a)$ for all $a \in A$. This function is called the* $\boxed{\text{restriction}}$ *of f to A and is denoted by* $\boxed{f|A}$.

Examples 8.1.12 (a) In Example 8.1.3 $f_1|\{a,b\} = f_2|\{a,b\}$ and is the constant function taking the value d.

(b) In Example 8.1.5 $f_2 = f_1|\mathbb{R}^{\geqslant}$ and $f_4 = f_3|\mathbb{R}^{\geqslant}$.

(c) In Example 8.1.6 $f_2|(\mathbb{R} - \{1\}) = f_4|(\mathbb{R} - \{1\}) = f_1$.

8.2 Composition of functions

Suppose that $f: X \to Y$ and $g: Y \to Z$. Then, given an element $x \in X$, the function f assigns to it an element $y = f(x) \in Y$ and now the function g assigns to this an element $g(y) = g\big(f(x)\big) \in Z$. Thus using f and g an element of Z has been assigned to x. This process defines a function with domain X and codomain Z called the composite of f and g.

Definition 8.2.1 *Given two functions $f: X \to Y$ and $g: Y \to Z$ the* $\boxed{\text{composite}}$ *of f and g, denoted by* $\boxed{g \circ f: X \to Z}$ *or simply $gf: X \to Z$, is defined by*

$$g \circ f(x) = g\big(f(x)\big) \quad \textit{for all } x \in X.$$

The order of the f and g in this definition should be carefully noted. Since we have adopted the convention (usual† in most areas of mathematics) of writing the symbol for the function on the left of the element

† But not used universally. In some branches of algebra it is quite common to write the value $f(x)$ as xf or even x^f.

where it is being evaluated (viz. $f(x)$) it is natural to denote the function obtained by applying f and then applying g by the symbol $g \circ f$ which presents them (from left to right) in the opposite order to their application. We can write the following to indicate this:

$$g \circ f: X \xrightarrow{f} Y \xrightarrow{g} Z.$$

Examples 8.2.2 (a) The function $x \mapsto (x+1)^2$ from $\mathbb{R}$ to $\mathbb{R}$ is the composite of the function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x + 1$ and the function $g: \mathbb{R} \to \mathbb{R}$ defined by $g(x) = x^2$ because

$$g \circ f(x) = g(x+1) = (x+1)^2 \quad \text{for all } x \in X.$$

Notice that the order does matter here because

$$f \circ g(x) = f(x^2) = x^2 + 1;$$

composition of functions is not in general *commutative.* It is not usually a good idea to refer simply to 'the composite of f and g' when both orders are possible since there is no well-established convention about which order these words indicate. It is best to use the symbolism $g \circ f$ or $f \circ g$ in order to be quite clear.

(b) Suppose that A is a subset of a set X. Then we may define a function $i: A \to X$ by $i(a) = a$ for all $a \in A$. This function is called the *inclusion function* of A into X. If now $f: X \to Y$ is a function, then the composite $f \circ i: A \to Y$ is equal to the restriction $f|A: A \to Y$ of f to A.

(c) Suppose that $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = x^2+1$ and $g: \mathbb{R}-\{0\} \to \mathbb{R}$ is defined by $g(x) = 1/x$. Then strictly speaking the composite $g \circ f$ is not defined since the codomain of f is not the same as the domain of g.

However, every value of f lies in the domain of g since $x^2 + 1 \geqslant 1 > 0$ for all real numbers x so that g can be evaluated at each value of f. In a case like this we can still attach a meaning to the composite $g \circ f$ as the function given by $g \circ f(x) = g(f(x))$. This gives the function $\mathbb{R} \to \mathbb{R}$ determined by $g \circ f(x) = 1/(x^2 + 1)$.

Proposition 8.2.3 *Suppose that $f: X \to Y$, $g: Y \to Z$ and $h: Z \to W$ are functions. Then*

(i) $(h \circ g) \circ f = h \circ (g \circ f): X \to Z$,

(ii) $f \circ I_X = f = I_Y \circ f: X \to Y$.

Proof These results are proved simply by evaluating the functions. For example, given $x \in X$, both the functions in (i) assign $h(g(f(x)))$ to this element and so the functions are equal. $\square$

Notice that the first part of Proposition 8.2.3 means that we can write $h \circ g \circ f$ without ambiguity. Composition of functions is *associative.*

8.3 Sequences

Definition 8.3.1 *A function $f : \mathbb{Z}^+ \to A$ is called a* $\boxed{\text{sequence}}$ *in the set A.*

The study of sequences in $\mathbb{R}$ or $\mathbb{C}$ is of great importance in advanced calculus and numerical analysis. Sequences may be defined by a formula, such as $n \mapsto n^2$, $n \mapsto 1/n$, $n \mapsto 2^n$, $n \mapsto (1 + 1/n)^n$ and $n \mapsto 1/n!$. But more interesting are those sequences defined inductively. We saw in Chapter 5 that strictly speaking formulae involving the exponent n or the factorial $n!$ really have to be defined inductively. Another example of an inductively defined sequence considered there was the Fibonacci sequence.

It is common in advanced calculus to solve numerical problems by generating a sequence of numbers x_n which provide increasingly good approximations to the solution as n increases – the sequence *converges* to a *limit* which is the required solution.†

One important reason for becoming adept at handling quantifiers is to be able to handle the definition of the limit of a sequence. To illustrate the ideas we introduce the idea of a null sequence, a sequence with limit 0.

Definition 8.3.2 *Given a sequence $f\colon \mathbb{Z}^+ \to \mathbb{R}$ of real numbers, we say that the sequence is* $\boxed{\text{null}}$*, written* $\boxed{\lim f = 0}$ *or* $\boxed{\lim_{n\to\infty} f(n) = 0}$*, when*

$$\forall \varepsilon \in \mathbb{R}^+, \exists N \in \mathbb{Z}^+, \forall n \in \mathbb{Z}^+ \ (n \geqslant N \Rightarrow |f(n)| < \varepsilon).$$

† A good example is the Newton–Raphson method for solving an equation, described in most books on advanced calculus (see for example G.B. Thomas and R.S. Finney, *Calculus and analytic geometry,* Addison-Wesley, Eighth edition 1992).

The use of ε for a general positive real number in this definition is traditional. This is not the place to investigate the full implications of this definition which involves *three* quantifiers! Here is a simple example of its use.

Example 8.3.3 The sequence $n \mapsto 1/\sqrt{n}$ is null.

Constructing a proof. We are required to prove that

$$\forall \varepsilon \in \mathbb{R}^+, \exists N \in \mathbb{Z}^+, \forall n \in \mathbb{Z}^+ \,(n \geqslant N \Rightarrow 1/\sqrt{n} < \varepsilon).$$

[Notice that $|1/\sqrt{n}| = 1/\sqrt{n}$.] So suppose we are given $\varepsilon \in \mathbb{R}^+$. Then to demonstrate the existence of the integer N we examine when $1/\sqrt{n} < \varepsilon$.

$$1/\sqrt{n} < \varepsilon \Leftrightarrow 1/n < \varepsilon^2 \Leftrightarrow n > 1/\varepsilon^2.$$

Hence so long as we choose a positive integer N such that $N > 1/\varepsilon^2$ we will have $n \geqslant N \Rightarrow n > 1/\varepsilon^2 \Rightarrow 1/\sqrt{n} < \varepsilon$ as required.

So for example, if $\varepsilon = 1$ then we must choose $N > 1$, if $\varepsilon = 1/2$ then we must choose $N > 4$, and if $\varepsilon = 1/100$ then $N > 10000$. The smaller the number ε the greater the number N is required to be. But whatever the positive real number ε is the condition $N > 1/\varepsilon^2$ tells us how large N has to be.

Proof Given a positive real number ε, $1/\sqrt{n} < \varepsilon$ if and only if $n > 1/\varepsilon^2$. Hence if we choose $N > 1/\varepsilon^2$ then $n \geqslant N$ implies that $n > 1/\varepsilon^2$ so that $1/\sqrt{n} < \varepsilon$ is required. □

This type of proof involving multiple quantifiers is quite elaborate and this is in fact an extremely simple example. More complicated examples are not considered in this book but are dealt with in detail in books on infinite sequences and series, advanced calculus and analysis.†

8.4 The image of a function

Given a function $f\colon X \to Y$, it is not necessary that every element of Y is a value of the function. For example the function $\mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^2$ does not have -1 as a value. Thus we can obtain a subset of Y by considering those elements which are values.

† See, for example, R. Haggerty, *Fundamentals of mathematical analysis*, Addison-Wesley, Second edition 1993.

Definition 8.4.1 *Given a function $f: X \to Y$, the subset of the codomain Y consisting of those elements which are values of f is called the* $\boxed{\text{image}}$ *of f and is denoted by* $\boxed{\mathrm{Im} f}$. *Thus*

$$\mathrm{Im} f = \{\, f(x) \mid x \in X \,\}.$$

This means that the function f provides a constructive definition of the set $\mathrm{Im} f$, as discussed in Chapter 6.

8.5 The graph of a function

Definition 8.5.1 *Suppose that $f: X \to Y$ is a function. Then we define the* $\boxed{\text{graph}}$ *of f to be the subset of the Cartesian product $X \times Y$ given by*

$$G_f = \{\, (x, y) \in X \times Y \mid y = f(x) \,\} = \{\, \big(x, f(x)\big) \mid x \in X \,\}.$$

Remarks 8.5.2 In the case that X and Y are subsets of $\mathbb{R}$ this is the usual idea of the graph of a function. Work in calculus makes it clear that the graph gives a great deal of information about the function and that it is useful to develop skill in drawing graphs.

Example 8.5.3 The graphs of the functions f_1 and f_2 of Example 8.1.3 are as follows.

Here we indicate the elements of the graph by solid dots (•).

The graph of the function $f: X \to Y$ is a particular example of a subset of the Cartesian product $X \times Y$ determined by a predicate: the predicate is $y = f(x)$. Not every subset of $X \times Y$ arises as the graph of a function. Each column $\{x_0\} \times Y$ must contain a single element, namely

$(x_0, f(x_0))$. However, the function f is determined by its graph G_f, for, given $x_0 \in X$, there is a unique element $(x, y) = (x_0, y_0) \in G_f$ such that $x = x_0$, and then $f(x_0) = y_0$.†

Exercises

8.1 Define functions f and $g: \mathbb{R}^2 \to \mathbb{R}$ by:

$$\begin{aligned} f(x,y) &= \frac{x+y}{2} + \frac{|x-y|}{2}; \\ g(x,y) &= \begin{cases} x & \text{if } x \geqslant y, \\ y & \text{if } x \leqslant y. \end{cases} \end{aligned}$$

Prove that the function g is well-defined. Prove that $f = g$.

8.2 Define functions f and $g: \mathbb{R} \to \mathbb{R}$ by $f(x) = x^3$ and $g(x) = 1 - x$. Find the functions (i) $f \circ f$, (ii) $f \circ g$, (iii) $g \circ f$, (iv) $g \circ g$.

List the elements of the set $\{ x \in \mathbb{R} \mid fg(x) = gf(x) \}$.

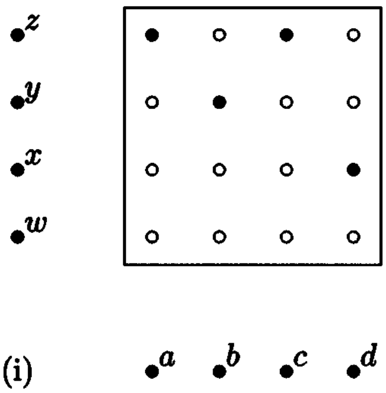
8.3 Find functions $f_i: \mathbb{R} \to \mathbb{R}$ with images as follows:

(i) $\mathrm{Im} f_1 = \mathbb{R}$;
(ii) $\mathrm{Im} f_2 = \mathbb{R}^+$;
(iii) $\mathrm{Im} f_3 = \mathbb{R} - \mathbb{Z}$;
(iv) $\mathrm{Im} f_4 = \mathbb{Z}$.

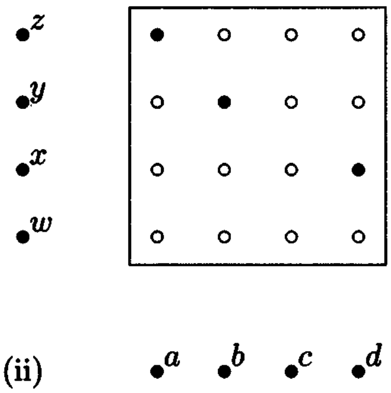
8.4 Prove that the sequence $n \mapsto 1/n$ is null.

8.5 Let $X = \{ a, b, c, d \}$ and $Y = \{ w, x, y, z \}$. Which of the following subsets of $X \times Y$ is the graph of a function $f: X \to Y$? For those which are write down a table for the corresponding function as in Example 8.1.3. Explain why the others are not graphs of functions.

† Our definition of a function (Definition 8.1.1) is usually considered rather informal in more advanced mathematics. There a function is defined to *be* a graph: in other words a function $X \to Y$ is defined to be a subset of $X \times Y$ in which each element of X occurs as the first coordinate of an element precisely once (see for example Daniel J. Velleman, *How to prove it, a structured approach*, Cambridge University Press, 1994).



(i)



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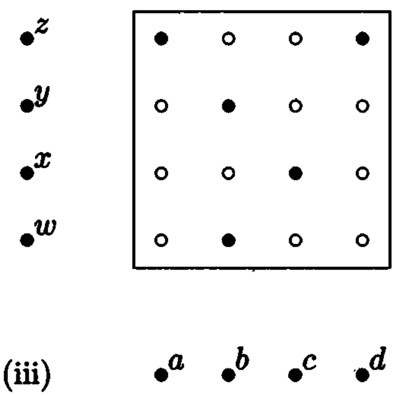

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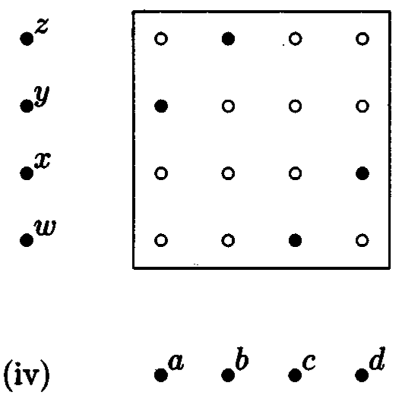


(iv)

9

Injections, surjections and bijections

In defining a function $f: X \to Y$ we insist that a unique element of Y is assigned to each element of X. However, we do not require that each element of Y is assigned to some element of X nor do we prevent the possibility of the same element of Y being assigned to several (or even all the) elements of X. By imposing additional conditions concerning the number of elements of X to which elements of Y are assigned we get functions with particular properties. In this chapter we consider functions with particularly good properties and in particular functions which are *bijections* for which we can define an inverse function.

9.1 Properties of functions

Definition 9.1.1 *Suppose that $f: X \to Y$ is a function.*

(i) *If no element of Y is assigned to more than one element of X, i.e. the function takes a different value for each point of the domain, then we say that the function f is an* $\boxed{\text{injection}}$ *(or that it is* injective *or* one-to-one*). In symbols we can write this*

$$\forall x_1, x_2 \in X, \ (x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2))$$

or equivalently using the contrapositive

$$\forall x_1, x_2 \in X, \ (f(x_1) = f(x_2) \Rightarrow x_1 = x_2).$$

(ii) *If each element of Y is assigned to some element of X, i.e. each point of the codomain is a value of the function, then we say that the* *function f is a* $\boxed{\text{surjection}}$ *(or that it is* surjective *or* onto*). In symbols we can write this*

$$\forall y \in Y, \exists x \in X, y = f(x).$$

(iii) *If f is both an injection and a surjection then we say that it is a* $\boxed{\text{bijection}}$ *(or that it is* bijective *or* one-to-one and onto*).*

We can reformulate these definitions using the notion of a *pre-image.*

Definition 9.1.2 *For a function $f: X \to Y$, given an element $y \in Y$ a* $\boxed{\text{pre-image}}$ *of y (under f) is an element $x \in X$ such that $y = f(x)$.*

So f is an injection if and only if every element of Y has at most one pre-image; f is a surjection if and only if every element of Y has at least one pre-image; and f is a bijection if and only if every element of Y has precisely one pre-image.

These ideas are simply expressed in terms of the various models for a function introduced in the previous chapter. If a function is described by listing the values of the function as in Example 8.1.2 then we find the pre-images of an element of the codomain by locating its occurrences in the list of values and reading up to the corresponding elements of the first row. Thus a function is injective when no element occurs more than once in the list of values, it is surjective when every element of the codomain does occur in the list of values and bijective when every element of the codomain occurs precisely once in the list of values.

In terms of the picture representing the function by arrows from the domain to the codomain (see page 90) we find pre-images by following arrows backwards. Thus, for example, a function is bijective if every point of the codomain is at the end of precisely one arrow.

Finally, if we think of a function as a procedure for placing the elements of the set X into a set of boxes Y (see page 90) then a pre-image of a particular box $y \in Y$ is simply an element which is placed in that box. From this point of view, f is injective when no two elements are placed in the same box, it is surjective when no box is left empty, and it is bijective when each box contains precisely one element.

Examples 9.1.3 (a) In Example 8.1.3 all the functions apart from the constant functions f_1 and f_8 are surjections. None of the functions are injections.

(b) In Example 8.1.4 the non-constant functions f_2 and f_3 are bijections. Notice that, for any set X, the identity function I_X is a bijection.

(c) In Example 8.1.5 the function f_1 is neither an injection (since $(-1)^2 = 1^2 = 1$) nor a surjection (since there is no real number x such that $x^2 = -1$); the function f_2 is an injection (since $0 \leqslant x_1 < x_2 \Rightarrow x_1^2 < x_2^2$) but not a surjection; the function f_3 is a surjection but not an injection; and the function f_4 is a bijection. This example illustrates the importance of the domain and the codomain. For example, if a result has been proved for injections then it can be applied to f_2 and to f_4 but not to f_1 and f_3.

Remarks 9.1.4 Notice that we can easily convert any function into a surjection by changing the codomain. Recall that, given a function $f: X \to Y$, the *image* of f, denoted by $\mathrm{Im} f$, is the set of values of f. The assignment determining the function f also determines a function $f^+: X \to \mathrm{Im} f$ (i.e. $f^+(x) = f(x)$) which is a surjection.

Since the definitions of injectivity and surjectivity involve universal and existential statements, proofs and disproofs of these properties usually follow the format discussed in Chapter 7.

Example 9.1.5 To determine whether the function $f_1: \mathbb{R} \to \mathbb{R}$ given by $f_1(x) = x + 1$ is an injection, a surjection or a bijection.

For injectivity, it is usually easiest to use the second formulation:

$$f_1(x_1) = f_1(x_2) \Rightarrow x_1 = x_2.$$

Filling in the definition of the function this gives

$$x_1 + 1 = x_2 + 1 \Rightarrow x_1 = x_2$$

which is certainly true so that f_1 is injective.

For surjectivity, we rewrite the definition as usual as a universal implication:

$$y \in \mathbb{R} \Rightarrow \exists x \in \mathbb{R},\, y = f_1(x).$$

If we put the definition of the function into this then it easy to prove the existential statement by example. For

$$y = f_1(x) \Leftrightarrow y = x + 1 \Leftrightarrow x = y - 1$$

which tells us that, given $y \in \mathbb{R}$, then $y = f_1(x)$ if $x = y - 1$ so that y does have a pre-image under f_1. Hence f_1 is surjective and so a bijection.

In fact, what is written above is rather inefficient; we can often prove bijectivity in one fell swoop. The best way to discover whether a function is injective or surjective can be to investigate the pre-images of a general element in the codomain using the approach used above for surjectivity. In this case if y is a general element of the codomain $\mathbb{R}$ then

$$\begin{aligned} x \text{ is a pre-image of } y \quad &\Leftrightarrow \quad y = f_1(x) \\ &\Leftrightarrow \quad y = x + 1 \\ &\Leftrightarrow \quad x = y - 1. \end{aligned}$$

This shows that each element y of $\mathbb{R}$ has precisely one pre-image, namely $y - 1$, so that the function is a bijection.

We can write out the solution to this problem quite briefly as follows.

Solution Since $y = f_1(x) \Leftrightarrow y = x + 1 \Leftrightarrow x = y - 1$, for $x, y \in \mathbb{R}$, we see that each element y of $\mathbb{R}$ has precisely one pre-image under f_1. Hence f_1 is a bijection. □

Example 9.1.6 To determine whether the function $f_2: \mathbb{R}^+ \to \mathbb{R}^+$ given by $f_2(x) = x + 1$ is injective, surjective or bijective.

This function is given by the same formula as the previous example. If we try the pre-image approach we get almost the same argument:

$$\begin{aligned} x \text{ is a pre-image of } y \quad &\Leftrightarrow \quad y = f_2(x) \\ &\Leftrightarrow \quad y = x + 1 \\ &\Leftrightarrow \quad x = y - 1. \end{aligned}$$

The difference becomes clear when we ask whether the formula for the pre-image makes sense. Remember that if x is to be a pre-image then it must certainly lie in the domain $\mathbb{R}^+$ (otherwise $f_2(x)$ isn't even defined). But, given $y \in \mathbb{R}^+$, $y - 1 \in \mathbb{R}^+$ if and only if $y - 1 > 0$ (by the definition of $\mathbb{R}^+$), i.e. $y > 1$. So we see that y has a pre-image if and only if $y > 1$. This proves that the image of f_2 is the set $\{\, y \in \mathbb{R}^+ \mid y > 1 \,\}$ and, since this is not the whole of $\mathbb{R}^+$, f_2 is not surjective. As usual, in the formal proof we give an explicit counterexample, an element of $\mathbb{R}^+$ which is not a value of the function.

However, the function is injective and the proof is just the same as in the previous example.

Solution The function is injective since, for $x_1, x_2 \in \mathbb{R}^+$, $f_2(x_1) = f_2(x_2) \Rightarrow x_1 + 1 = x_2 + 1 \Rightarrow x_1 = x_2$. However, if $x \in \mathbb{R}^+$ then $x > 0$ so that $f_2(x) = x + 1 > 1$. Thus $1/2$ is not a value of f_2 which is not then surjective. □

Example 9.1.7 To determine whether the function $f_3 : \mathbb{R} \to \mathbb{R}$ given by $f_3(x) = 4x^2 - 4x + 2$ is injective, surjective or bijective.

This time the pre-image approach gives

$$\begin{aligned} y = f_3(x) &\Leftrightarrow y = 4x^2 - 4x + 2 \\ &\Leftrightarrow y = (2x-1)^2 + 1 \\ &\Leftrightarrow x = (1 \pm \sqrt{y-1})/2. \end{aligned}$$

Again we must check whether our formula for the pre-image makes sense. Now if $y < 1$ then $y - 1$ has no (real) square root and so in that case the formula has no meaning. Looking back through the steps above we see what the problem is: $y = (2x - 1)^2 + 1$ tells us that if y is a value then $y \geqslant 1$. This tells us that f_3 is not surjective.

The other thing we notice in the formula for the pre-image is the symbol $\pm$. This means that x is a pre-image for y if and only if $x = (1 + \sqrt{y-1})/2$ or $x = (1 - \sqrt{y-1})/2$. In other words if $\sqrt{y-1}$ exists and is non-zero, then y has two pre-images. For example, taking $y = 2$ for which $\sqrt{y-1} = 1$,† we obtain $f_3(x) = 2 \Leftrightarrow x = (1 \pm 1)/2$, i.e. $x = 1$ or $x = 0$. So f_3 is not injective.

Solution Because $f_3(0) = f_3(1) = 2$, the function is not injective.

For a real number x, $f_3(x) = 4x^2 - 4x + 2 = (2x - 1)^2 + 1 \geqslant 1$. Thus 0 is not a value of f_3 and so it is not surjective. □

Notice that in presenting this proof no indication is given of how we found the counterexample to injectivity. Formally this is not required for the proof. In a case like this it is not difficult to spot such a counterexample (for example from the graph of the function or from rewriting the formula as $(2x - 1)^2 + 1$) but it is often a good idea to help the reader by indicating how you found the counterexample.

† We adopt the usual convention that if y is a non-negative real number then $\sqrt{y}$ represents the non-negative square root of y, i.e. $x = \sqrt{y}$ if and only if $y = x^2$ and $x \geqslant 0$.

Example 9.1.8 To determine whether the function $f_4: \mathbb{R}^+ \to \{ x \in \mathbb{R}^+ \mid x \geqslant 1 \}$ given by $f_4(x) = 4x^2 - 4x + 2$ is injective, surjective or bijective.

The pre-image approach gives just the same formula as in the previous example. However, if y is in the codomain then $y \geqslant 1$ and so the square root in the formula for a pre-image element does determine a real number. Furthermore $(1 + \sqrt{y-1})/2 > 1/2$ and so this gives one pre-image for y showing that f_4 is surjective. For $y > 1$, we have a second pre-image precisely when $(1 - \sqrt{y-1})/2 > 0$ which occurs when $\sqrt{y-1} < 1$, i.e. $y < 2$. This proves that f_4 is not injective since elements y in the codomain have two pre-images if $y < 2$. However, it is probably more satisfying to give two specific elements of the domain where the function takes the same value; we can get these by taking $y = 5/4$ which has $\sqrt{y-1} = 1/2$ and so the pre-images $(1 \pm 1/2)/2$, i.e. $3/4$ and $1/4$.

Sketching the graph of this function is a great aid to seeing what is going on. The reader should do this.

Solution Notice that $y = f_4(x) \Leftrightarrow x = (1 \pm \sqrt{y-1})/2$. The function is surjective since given $y \geqslant 1$, $y = f_4(x)$ for $x = (1 + \sqrt{y-1})/2 \in \mathbb{R}^+$. The function is not injective since $f_4(1/4) = f_4(3/4) = 5/4$. $\square$

9.2 Bijections and inverses

Definition 9.2.1 *A function $f: X \to Y$ is called* $\boxed{\text{invertible}}$ *iff*† *there exists a function $g: Y \to X$ such that*

$$y = f(x) \Leftrightarrow x = g(y) \quad \textit{for all } x \in X \textit{ and for all } y \in Y.$$

In this case we call g an $\boxed{\text{inverse (function)}}$ *of f and write* $\boxed{g = f^{-1}}$.

The symmetry of the definition shows that in this case g is also invertible and f is an inverse of g.

This means then that corresponding to the assignment of an element of Y to each element of X there is a reverse assignment of an element of X to each element of Y.

† The reader should note the way that the word 'if' is used in this definition. It really means 'if and only if' since we are defining the meaning of the word 'invertible'. Saying that f is invertible means precisely the same as saying that there is a function g with the properties given. This usage in definitions is very common although it has on the whole been avoided so far in this book.

Example 9.2.2 Suppose that $f: \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = 2x + 1$ and $g: \mathbb{R} \to \mathbb{R}$ is defined by $g(x) = (x - 1)/2$. Then, for $x, y \in \mathbb{R}$,

$$y = f(x) \Leftrightarrow y = 2x + 1 \Leftrightarrow x = (y - 1)/2 \Leftrightarrow x = g(y).$$

Thus f and g are invertible and each is the inverse of the other.

Theorem 9.2.3 *Let $f: X \to Y$. Then f is invertible if and only if it is a bijection. Furthermore, if it is invertible, then its inverse function is unique.*

Constructing a proof. Intuitively this result is pretty clear. For example if we think of the model of the function given by placing elements from the set X in the set of boxes Y then we have observed that if the function is bijective there is precisely one element of X in each box. In this case the inverse function is given by associating to each box the element which it contains. Furthermore this procedure is only possible if there is only one element in each box which means that f is bijective.

The formal proof makes these ideas a bit more precise in terms of the formal definitions of bijectivity and inverse and is constructed as so often by simply spelling out these definitions. It may seem quite long, but if you are clear about what you are trying to do there is really only one possible way forward at each stage and so the proof more or less writes itself.

The last part of the theorem asserts that f has a unique inverse. This means that f has an inverse (for this is what is meant by stating that it is invertible) but only one inverse. Uniqueness statements are very common. The usual approach to proving them is illustrated in this case: we demonstrate that if g_1 and g_2 are inverses of f then $g_1 = g_2$. Arguments of this type have already been met in proofs of the injectivity of a function f, for this asserts that each value of the function has a unique pre-image: this is proved by demonstrating that if x_1 and x_2 are pre-images of the same element then $x_1 = x_2$ (see for example Example 9.1.5).

Proof For the first statement two things have to be proved.

(a) 'f invertible $\Rightarrow$ f bijective': Suppose that f is invertible. Then, by definition, there is an inverse function $g: Y \to X$ so that

$$y = f(x) \Leftrightarrow x = g(y) \quad \text{for all } x \in X \text{ and for all } y \in Y.$$

Now to prove that f is bijective we must prove that f is surjective and that it is injective.

For injectivity, suppose that $x_1, x_2 \in X$ are such that $f(x_1) = f(x_2)$; then we must prove that $x_1 = x_2$. Put $y_0 = f(x_1) = f(x_2)$. Then $y_0 = f(x_1) \Rightarrow x_1 = g(y_0)$ (from the definition of the inverse) but on the other hand $y_0 = g(x_2) \Rightarrow x_2 = g(y_0)$. Hence $x_1 = g(y_0) = x_2$ as required.

For surjectivity suppose that $y_0 \in Y$. We must demonstrate that there exists an element $x_0 \in X$ such that $y_0 = f(x_0)$. Put $x_0 = g(y_0)$. Then it is immediate from the definition of an inverse function that $y_0 = f(x_0)$ as required.

(b) 'f bijective $\Rightarrow$ f invertible': Suppose that f is bijective. To show that it is invertible we must construct an inverse function $g: Y \to X$. Suppose that $y_0 \in Y$. Since f is a bijection, y_0 has precisely one pre-image, say $x_0 \in X$, such that $f(x_0) = y_0$. So we can define a function by the rule that, for each $y \in Y$, $g(y)$ is the unique element $x \in X$ such that $f(x) = y$. It follows from this definition that

$$y = f(x) \Leftrightarrow x = g(y) \quad \text{for all } x \in X \text{ and for all } y \in Y$$

and so g is an inverse of f.

(c) 'f invertible $\Rightarrow$ it has a unique inverse': Suppose that f is invertible and $g_1: Y \to X$ and $g_2: Y \to X$ are inverse functions for f. To demonstrate that $g_1 = g_2$ we must show that $g_1(y) = g_2(y)$ for all $y \in Y$. Let $y_0 \in Y$. Then put $x_1 = g_1(y_0)$ and $x_2 = g_2(y_0)$ so that we are required to prove that $x_1 = x_2$. This is easy because $x_1 = g_1(y_0) \Rightarrow y_0 = f(x_1)$ and $x_2 = g_2(y_0) \Rightarrow y_0 = f(x_2)$ so that $f(x_1) = f(x_2)$. But now, since f is invertible it is bijective (by (a)) and so in particular injective. It follows that $x_1 = x_2$, i.e. $g_(y_0) = g_2(y_0)$ as required. □

Because of this theorem the words 'bijective' and 'invertible' are used interchangeably when referring to functions. Many standard functions are not actually bijections but by suitable restrictions of the domain and codomain can be converted into bijections. Here are some examples from calculus.

Examples 9.2.4 (a) The function $\sin: \mathbb{R} \to \mathbb{R}$ is neither injective (since, for example, $\sin 0 = \sin \pi$) nor surjective (since, for example, there is no real number x such that $\sin x = 2$). To define an inverse function $\sin^{-1}$ we restrict the domain and the codomain of sin so that it is bijective. Surjectivity is easy: we simply change the codomain to the image of sin, i.e. Im(sin), which is the set $[-1,1] = \{x \in \mathbb{R} \mid -1 \leqslant x \leqslant 1\}$ (since we know that each number in this set arises as a value of the sine function on $\mathbb{R}$). Injectivity is not much harder for we know that as x varies from $-\pi/2$ to $\pi/2$ the value of $\sin x$ increases steadily from -1 to 1 (sketch the graph). So we take as our domain $[-\pi/2, \pi/2] = \{x \in \mathbb{R} \mid -\pi/2 \leqslant x \leqslant \pi/2\}$. To sum up, the function†

$$\sin\colon [-\pi/2, \pi/2] \to [-1,1]$$

is a bijection. By the above theorem it has an inverse

$$\sin^{-1}\colon [-1,1] \to [-\pi/2, \pi/2].$$

In making this definition we had to choose a subset of $\mathbb{R}$ on which the sine function was injective. There are many possibilities, for example $[\pi/2, 3\pi/2]$, and each has its own inverse function. The above choice is the most natural and the function it leads to is called the *principal value* of the inverse.

In the same way we can restrict the cosine and tangent functions to obtain inverse functions

$$\cos^{-1}\colon [-1,1] \to [0,\pi]$$

and

$$\tan^{-1}\colon \mathbb{R} \to (-\pi/2, \pi/2).$$

(b) Given $n \in \mathbb{Z}^+$, the function $x \mapsto x^n$ is a bijection $\mathbb{R} \to \mathbb{R}$ for n odd and so in this case we do have an inverse function $\mathbb{R} \to \mathbb{R}$, the *nth root function*, denoted by‡ $x \mapsto x^{1/n}$ or $x \mapsto \sqrt[n]{x}$.

† Strictly speaking this function should have a different name in order to indicate the change of domain and codomain. However, it is usual to use the same name.

‡ The notation $x^{1/n}$ is selected so that the law of exponents holds: $(x^{1/n})^n = (x^n)^{1/n} = x^1 = x$. With this definition it is possible to give some indication of the problem in defining 0^0 referred to in Definition 5.3.3. Notice that $0^{1/n}$ is defined to be 0. Furthermore, we have observed (Exercise 8.4) that the sequence $1/n$ is null, $\lim 1/n = 0$, which suggests that we ought to define $0^0 = 0$. However, if the law of indices is to hold then we must have $x^0 = 1$ for non-zero x (see the solution to Exercise 5.7) and this suggests that $0^0 = 1$, the convention adopted in this book. It is not possible to satisfy these conflicting demands. For $(x,y) \in \mathbb{R}^{\geqslant} \times \mathbb{R} - \{(0,0)\}$ it is possible to define x^y extending the definition for integer exponenents so that the laws of exponents are satisfied and so that the function of two variables $(x,y) \mapsto x^y$ is 'continuous'. It is not possible to extend this 'continuous' function to the point $(0,0)$. A full explanation of this, and the definition of 'continuous', is beyond the scope of this book (see for example K.G. Binmore, *Mathematical analysis, a straightforward approach,* Cambridge University Press, Second edition 1982).

On the other hand it is neither an injection nor a surjection when n is even. [The case $n = 2$ is typical and is discussed in Example 9.1.3(c).] In this case we restrict the domain and the codomain to $\mathbb{R}^{\geqslant}$ so that we have a bijection. Thus the inverse function $x \mapsto x^{1/n} = \sqrt[n]{x}$ is a function $\mathbb{R}^{\geqslant} \to \mathbb{R}^{\geqslant}$. The convention is that for even n the expressions $x^{1/n}$ and $\sqrt[n]{x}$ represent the non-negative root of a non-negative real number x.

The following equivalent formulation of Definition 9.2.1 is often useful.

Proposition 9.2.5 *The functions $f: X \to Y$ and $g: Y \to X$ are inverses of each other if and only if $g \circ f = I_X$ and $f \circ g = I_Y$.*

Proof '$\Rightarrow$': Suppose that f and g are inverse to each other. Then

$$y = f(x) \Leftrightarrow x = g(y) \quad \text{for all } x \in X \text{ and all } y \in Y.$$

Now, given $x_0 \in X$, put $y_0 = f(x_0)$ so that $x_0 = g(y_0)$. Then $g \circ f(x_0) = g(y_0) = x_0$, i.e. $g \circ f = I_X$.

Similarly $f \circ g = I_Y$.

'$\Leftarrow$': Suppose that $g \circ f = I_X$ and $f \circ g = I_Y$. Suppose that $y_0 = f(x_0)$. Then $g(y_0) = g \circ f(x_0) = I_X(x_0) = x_0$. Thus $y = f(x) \Rightarrow x = g(y)$ for all $x \in X$, $y \in Y$.

Similarly, $x = g(y) \Rightarrow y = f(x)$ and so we have proved that f and g are inverses of each other. □

9.3 Functions and subsets

In the previous section it has been demonstrated that a function f has an inverse f^{-1} if and only if it is a bijection. However, the reader will discover that the notation f^{-1} which has been used for the inverse function is used even when the function f is not a bijection. This chapter ends with a brief description of this more general use.

To begin with recall the definition of the power set (Definition 6.3.1): the power set $\mathcal{P}(X)$ of a set X is the set of subsets of X so that $A \in \mathcal{P}(X)$ means simply $A \subseteq X$. We now describe how given a function $f: X \to Y$ we can associate to it two functions between the power sets of X and Y (one in each direction) each of which captures the function f in a different way.

Definition 9.3.1 *Suppose that $f: X \to Y$ is a function.*

(i) *The function* $\overrightarrow{f}: \mathcal{P}(X) \to \mathcal{P}(Y)$ *is defined by*

$$\overrightarrow{f}(A) = \{ f(x) \mid x \in A \}$$

for $A \in \mathcal{P}(X)$.

(ii) *The function* $\overleftarrow{f}: \mathcal{P}(Y) \to \mathcal{P}(X)$ *is defined by*

$$\overleftarrow{f}(B) = \{ x \in X \mid f(x) \in B \}$$

for $B \in \mathcal{P}(Y)$.

Remarks 9.3.2 The notations $\overrightarrow{f}$ and $\overleftarrow{f}$ are non-standard. Most writers denote the function $\overrightarrow{f}$ simply by f and denote the function $\overleftarrow{f}$ by f^{-1}. With experience this does not lead to confusion (and is another example of how ambiguity can be better than pedantry). However, in this book the notations $\overrightarrow{f}$ and $\overleftarrow{f}$ will be used since it seems potentially confusing to have two different functions with the same name when meeting these ideas for the first time.

In the case of $\overrightarrow{f}$ notice that each element x_0 of X corresponds to an element of $\mathcal{P}(X)$, namely the singleton subset $\{x_0\}$, and similarly for Y. So we can think of $\overrightarrow{f}$ as an extension of f in the sense that

$$\overrightarrow{f}(\{x_0\}) = \{ f(x) \mid x \in \{x_0\}\} = \{ f(x) \mid x = x_0 \} = \{f(x_0)\}.$$

Notice that $\overrightarrow{f}(X)$ is the image of f, $\mathrm{Im}(f)$ (see Definition 8.4.1).

In the case of $\overleftarrow{f}$,

$$\overleftarrow{f}(\{y_0\}) = \{ x \in X \mid f(x) \in \{y_0\}\} = \{ x \in X \mid f(x) = y_0 \}$$

for each $y_0 \in Y$. Thus $\overleftarrow{f}(\{y_0\})$ gives the set of pre-images of y_0. In the box model of a function (page 90) this gives the set of elements in the box y_0. Now, if f is a bijection with inverse f^{-1}, then $f(x) = y_0$ if and only if $x = f^{-1}(y_0)$ so that $\overleftarrow{f}(\{y_0\}) = \{f^{-1}(y_0)\}$ and we can think of $\overleftarrow{f}$ as an extension of f^{-1}.

If f is not a bijection then $\overleftarrow{f}(\{y\})$ will not be a singleton subset for some elements of Y: if it is not a surjection then it will be empty for elements y not in the image, and if not an injection then it will contain more than one element for some y.

9.4 Peano's axioms for the natural numbers

The notion of the successor of an integer n was introduced in Section 5.1. Chapter 5 was headed by a quotation from Richard Dedekind suggesting that this idea captured the essence of the natural numbers (or positive integers). The necessary language is now available so that some indication of the ideas behind the Dedekind quotation can be given.

Definition 9.4.1 *The* $\boxed{\text{successor function}}$, $\boxed{s\colon \mathbb{Z}^+ \to \mathbb{Z}^+}$, *is defined by* $s(n) = n+1$ *for* $n \in \mathbb{Z}^+$.

Dedekind observed that the existence of the successor function together with the number 1 completely captures the properties of the natural numbers. These ideas are now usually associated with the name of the Italian mathematician Giuseppe Peano.†

Axioms 9.4.2 (Peano) *The set of positive integers* $\mathbb{Z}^+$ *is a set with a function* $s : \mathbb{Z}^+ \to \mathbb{Z}^+$ *and an element* $1 \in \mathbb{Z}^+$ *such that*

(i) *s is an injection,*
(ii) *1 is not in the image of s, and*
(iii) *for $A \subseteq \mathbb{Z}^+$, if $1 \in A$, and $n \in A \Rightarrow s(n) \in A$, then $A = \mathbb{Z}^+$.*

Notice that (iii) is just the induction axiom as reformulated in Axiom 7.5.1.

† The first exposition of these ideas was by Richard Dedekind in his book *Was sind und was sollen die Zahlen?* published in 1888. Peano formulated his axioms for the natural numbers in his book *Arithmetices principia, nova methodo exposita* published in 1889. Peano did not see Dedekind's work until his own book was in press and Peano's exposition using the language of set theory is considered to have been much more influential. It was in this book that he introduced notations for set membership (the modern symbol $\in$) and set inclusion (he used an inverted 'C' which he also used for implication – recall that it was observed in Section 6.1 that there is a strong connection between set inclusion and implication) and he seems to have been the first to clarify the distinction between set membership and set inclusion. Peano was a remarkable mathematician. It is curious that his 1889 book was written in Latin rather than the Italian or French he usually used. In his book about Peano (Reidel, 1980), Hubert Kennedy describes it as 'the unique romantic act of his scientific career' and suggests that it reflected Peano's awareness that he was doing something historic – after all, the great classics such as Isaac Newton's *Principia* had been written in Latin. The following year brought Peano's most striking achievement, his construction of a space-filling curve; this is a curve in the plane $\mathbb{R}^2$ given by continuous functions $x = f(t)$, $y = g(t)$ such that, as t varies over the interval $[0,1]$, (x,y) passes through every point of the unit square $[0,1] \times [0,1]$. This was perhaps the first indication of the subtlety of the notion of dimension, confounding intuition, whose exploration has now led to the modern study of fractal sets. (See for example Donald M. Davis, *The nature and power of mathematics*, Princeton University Press, 1993.)

It is not difficult to prove (by induction) that, given any set X with a function $s : X \to X$ and a distinguished element $1 \in X$ satisfying these axioms, there is a bijection $X \to \mathbb{Z}^+$ under which the distinguished elements 1 and the successor functions correspond.† This is what is meant by saying that these statements provide axioms for the positive integers.

We can use s to define algebraic operations on $\mathbb{Z}^+$ inductively as follows.

Definition 9.4.3 (Peano) *The sum $m + n$ of positive integers m and n may be defined by induction on n by*

(i) $m + 1 = s(m)$, *and*
(ii) *for* $k \in \mathbb{Z}^+$, $m + s(k) = s(m + k)$.

The product of positive integers mn or $m \times n$ is now defined (making use of addition) by induction on n by

(i) $m \times 1 = m$, *and*
(ii) *for* $k \in \mathbb{Z}^+$, $m \times s(k) = m \times k + m$.

It is an interesting exercise to prove the basic algebraic properties of addition and multiplication starting from these definitions, for example the commutativity of addition $m + n = n + m$.

Exercises

9.1 Determine whether each of the following functions $\mathbb{R} \to \mathbb{R}$ is injective, surjective or bijective.

(i) $f_1(x) = 2x + 5$.
(ii) $f_2(x) = x^2 + 2x + 1$.
(iii) $f_3(x) = x^2 - 2x$.
(iv) $f_4(x) = \begin{cases} 1/x & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$

9.2 In each of the above examples consider the effect of changing the domain and the codomain to $\mathbb{R}^+$.

† See for example G. Birkhoff and S. Mac Lane, *A survey of modern algebra*, Macmillan, Fourth edition 1977.

9.3 Find inverses for the following functions:

(i) $f_1: \mathbb{R} \to \mathbb{R}$ given by $f_1(x) = 3x + 2$;
(ii) $f_2: \mathbb{R} \to \mathbb{R}$ given by $f_2(x) = x^3 + 1$.

9.4 Suppose that $f: X \to Y$ and $g: Y \to Z$ are injections. Prove that $g \circ f: X \to Z$ is an injection.

9.5 Suppose that $f: X \to Y$ and $g: Y \to Z$ are bijections of sets. Prove that the composite $g \circ f: X \to Z$ is also a bijection and that

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}: Z \to Y \to X.$$

9.6 Let $f: X \to Y$ be a function with graph $G_f \subseteq X \times Y$. Prove that f is surjective if and only if $\forall y \in Y$, $(X \times \{y\} \cap G_f) \neq \emptyset$.

9.7 Let $f: X \to Y$ be a function and $B_1, B_2 \in \mathcal{P}(Y)$. Prove that

(i) $B_1 \subseteq B_2 \Rightarrow \overleftarrow{f}(B_1) \subseteq \overleftarrow{f}(B_2)$,
(ii) $\overleftarrow{f}(B_1 \cap B_2) = \overleftarrow{f}(B_1) \cap \overleftarrow{f}(B_2)$,
(iii) $\overleftarrow{f}(B_1 \cup B_2) = \overleftarrow{f}(B_1) \cup \overleftarrow{f}(B_2)$.

Prove that the converse of the first of these statements is not universally true (by constructing a counterexample).

Problems II: Sets and functions

1 Prove the following statements:

(i) $\{x \in \mathbb{R} \mid x^2 - 3x + 2 = 0\} = \{x \in \mathbb{Z} \mid 0 < x < 3\}$;
(ii) $\{x \in \mathbb{R} \mid x^2 - 3x + 2 < 0\} = \{x \in \mathbb{R} \mid x < 2\} \cap \{x \in \mathbb{R} \mid x > 1\}$;
(iii) $\{x \in \mathbb{R} \mid x^2 - 3x + 2 > 0\} = \{x \in \mathbb{R} \mid x > 2\} \cup \{x \in \mathbb{R} \mid x < 1\}$.

2. By using a truth table prove that, for sets A, B and C,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Draw a Venn diagram to illustrate the proof.

3. Prove the absorption laws:

(i) $A \cap (A \cup B) = A$;
(ii) $A \cup (A \cap B) = A$.

4. Prove by contradiction or otherwise that $A \cap B = A \cap C$ and $A \cup B = A \cup C$ if and only if $B = C$.

5. Using truth tables, prove that for sets A, B and C,

(i) $(A \cup C) - B \subseteq (A - B) \cup C$,
(ii) $(A \cap C) - B = (A - B) \cap C$.

Draw Venn diagrams to illustrate the proofs.

Prove that there is equality in the first of these results if and only if $B \cap C = \emptyset$.

Deduce from the second of these results that

$$(A - B) \cap C = \emptyset \text{ if and only if } A \cap C \subseteq B.$$

6. Use the distributivity law to prove that

$$(A \cap B) \cup (B \cap C) \cup (C \cap A) = (A \cup B) \cap (B \cup C) \cap (C \cup A).$$

7. For subsets of a universal set U prove that $B \subseteq A^c$ if and only if $A \cap B = \emptyset$. By taking complements deduce that $A^c \subseteq B$ if and only if $A \cup B = U$. Deduce that $B = A^c$ if and only if $A \cap B = \emptyset$ and $A \cup B = U$.

8. Given sets $A, B \in \mathcal{P}(X)$, their *symmetric difference* is defined by

$$A \Delta B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B).$$

Prove that

(i) the symmetric difference is associative, $(A \Delta B) \Delta C = A \Delta (B \Delta C)$ for all $A, B, C \in \mathcal{P}(X)$,
(ii) there exists a unique set $N \in \mathcal{P}(X)$ such that $A \Delta N = A$ for all $A \in \mathcal{P}(X)$
[Hint: Guess what N is!],
(iii) for each $A \in \mathcal{P}(X)$, there exists a unique $A' \in \mathcal{P}(X)$ such that $A \Delta A' = N$,
(iv) for each $A, B \in \mathcal{P}(X)$, there exists a unique set C such that $A \Delta C = B$.

9. Using the notation of the previous problem, prove that for sets A, B, C, $D \in \mathcal{P}(X)$

$$A \Delta B = C \Delta D \Leftrightarrow A \Delta C = B \Delta D.$$

10. We define half-infinite† intervals as follows:

$$\begin{aligned}(a, \infty) &= \{x \in \mathbb{R} \mid x > a\}; \\ [a, \infty) &= \{x \in \mathbb{R} \mid x \geqslant a\}.\end{aligned}$$

Prove that

(i) $(a, \infty) \subseteq [b, \infty) \Leftrightarrow a \geqslant b$,
(ii) $[a, \infty) \subseteq (b, \infty) \Leftrightarrow a > b$.

† The symbol '∞' used in the notation in this question does not represent a number 'infinity'. The expression (a, ∞) is used by analogy with (a, b), where b is a real number, but the definitions are different (cf. Exercise 6.1). In this book no meaning is attached to the symbol '∞' on its own and it is only used in the notations in this question and in Definition 8.3.2. Further discussion of this use of the symbol can be found in books on analysis, for example R. Haggerty, *Fundamentals of mathematical analysis,* Addison-Wesley, Second edition 1993.

11. Give a proof or a counterexample for each of the following statements:

(i) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y > 0$;
(ii) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x - y > 0$;
(iii) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x + y > 0$;
(iv) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, xy > 0$;
(v) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, xy > 0$;
(vi) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, xy \geqslant 0$;
(vii) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, xy \geqslant 0$;
(viii) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, (x + y > 0 \text{ or } x + y = 0)$;
(ix) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, (x + y > 0 \text{ and } x + y = 0)$;
(x) $(\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y > 0)$ and $(\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x + y = 0)$.

12. Suppose that $A \subseteq \mathbb{Z}$. Write the following statement entirely in symbols using the quantifiers $\forall$ and $\exists$. Write out the negative of this statement in symbols.

There is a greatest number in the set A.

Give an example of a set A for which this statement is true. Give another example for which it is false.

13. Prove that, for sets A, B, C and D,

(i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$,
(ii) $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$,

14. Define functions f and $g: \mathbb{R} \to \mathbb{R}$ by $f(x) = x^2$ and $g(x) = x^2 - 1$. Find the functions $f \circ f$, $f \circ g$, $g \circ f$, $g \circ g$.

List the elements of the set $\{\, x \in \mathbb{R} \mid fg(x) = gf(x) \,\}$.

15. Given $A \in \mathcal{P}(X)$ define the *characteristic function* $\chi_A: X \to \{0, 1\}$ by

$$\chi_A(x) = \begin{cases} 0 & \text{if } x \notin A, \\ 1 & \text{if } x \in A. \end{cases}$$

Suppose that A and B are subsets of X.

(i) Prove that the function $x \mapsto \chi_A(x)\chi_B(x)$ (multiplication of integers) is the characteristic function of the intersection $A \cap B$.
(ii) Find the subset C whose characteristic function is given by

$$\chi_C(x) = \chi_A(x) + \chi_B(x) - \chi_A(x)\chi_B(x).$$

16. Determine which of the following functions $f_i: \mathbb{R} \to \mathbb{R}$ are injective, which are surjective and which are bijective. Write down an inverse function of each of the bijections.

(i) $f_1(x) = x - 1$;
(ii) $f_2(x) = x^3$;
(iii) $f_3(x) = x^3 - x$;
(iv) $f_4(x) = x^3 - 3x^2 + 3x - 1$;
(v) $f_5(x) = e^x$;
(vi) $f_6(x) = \begin{cases} x^2 & \text{if } x \geqslant 0, \\ -x^2 & \text{if } x \leqslant 0. \end{cases}$

17. Functions $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ are defined as follows.

$$f(x) = \begin{cases} x+2 & \text{if } x < -1, \\ -x & \text{if } -1 \leqslant x \leqslant 1, \\ x-2 & \text{if } x > 1. \end{cases}$$

$$g(x) = \begin{cases} x-2 & \text{if } x < -1, \\ -x & \text{if } -1 \leqslant x \leqslant 1, \\ x+2 & \text{if } x > 1. \end{cases}$$

Find the functions $f \circ g$ and $g \circ f$. Is g the inverse of the function f? Is f injective or surjective? How about g? Sketch and compare the graphs of these functions.

18. Suppose that $f: X \to Y$ and $g: Y \to Z$ are surjections. Prove that the composite $g \circ f: X \to Z$ is a surjection.

19. Let $f: X \to Y$ be a function. Prove that there exists a function $g: Y \to X$ such that $f \circ g = I_Y$ if and only if f is a surjection. [g is called a *right inverse* of f.]

20. Let $f: X \to Y$ be a function and $A_1, A_2 \in \mathcal{P}(X)$.

(i) Prove that $A_1 \subseteq A_2 \Rightarrow \overrightarrow{f}(A_1) \subseteq \overrightarrow{f}(A_2)$. Prove that the converse is not universally true. Give a simple condition on f which is equivalent to the converse.
(ii) Prove that $\overrightarrow{f}(A_1 \cap A_2) \subseteq \overrightarrow{f}(A_1) \cap \overrightarrow{f}(A_2)$. Prove that equality is not universally true.
(iii) Prove that $\overrightarrow{f}(A_1 \cup A_2) = \overrightarrow{f}(A_1) \cup \overrightarrow{f}(A_2)$.

21. Let $f: X \to Y$ be a function. Prove that

(i) f is injective $\Leftrightarrow \overrightarrow{f}$ is injective $\Leftrightarrow \overleftarrow{f}$ is surjective,

(ii) f is surjective $\Leftrightarrow \overrightarrow{f}$ is surjective $\Leftrightarrow \overleftarrow{f}$ is injective.

22. Starting from Peano's axioms prove that if $n \in \mathbb{Z}^+$ and $n \neq 1$ then n is a successor, i.e. $s(a) = n$ for some $a \in \mathbb{Z}^+$.

[Let $A = \mathrm{Im}(s) \cup \{1\}$ and prove that $A = \mathbb{Z}^+$.]

23. Starting from Definition 9.4.3 prove that addition of positive integers is

(i) associative, i.e. $(a + b) + c = a + (b + c)$ for positive integers a, b, c,

(ii) commutative, i.e. $a + b = b + a$ for positive integers a, b.

[For (ii), prove first of all that $a + 1 = 1 + a$ by induction on a.]

Part III

Numbers and counting

10
Counting

One, two, buckle my shoe;
Three, four, knock at the door;
Five, six, pick up sticks;
Seven, eight, lay them straight;
Nine, ten, a big fat hen.
Eleven, twelve, dig and delve;
Thirteen, fourteen, maids a-courting;
Fifteen, sixteen, maids a-kissing;
Seventeen, eighteen, maids a-waiting;
Nineteen, twenty, my plate's empty.
Traditional.

Perhaps the oldest mathematical activity is counting. Some of the earliest words we learn as children are the names of the first few natural numbers and counting games are very common. In this chapter the counting process is formulated mathematically as the constructing of a bijection from a standard set of positive integers to the set of the elements being counted. This allows us to consider precisely what we mean by the number of elements in a set and what it means for a set to be finite.

10.1 Counting finite sets

Recall that we have suggested that a set can be thought of as a box containing the elements of the set. The process of counting the elements in the set or box can be carried out by removing them in turn one by one while reciting the names of the positive integers in order: 'one', 'two', 'three', and so on. When we have exhausted the elements, we say that the last number recited is the number of elements in the set, or the *cardinality* of the set. Sometimes, instead of removing the elements we simply point to them in turn as when we count the number of people in a room, but the idea is the same. What this amounts to is the construction of a bijection from the standard set

$$\mathbb{N}_n = \{1, 2, \ldots, n\} = \{\, i \in \mathbb{Z} \mid 1 \leqslant i \leqslant n \,\},$$

the set of the first n positive integers for some positive integer n, to the set being counted. As we count a set X, the counting process effectively defines a bijection $f\colon \mathbb{N}_n \to X$ by setting $f(1)$ equal to the first element of X which we count, $f(2)$ equal to the second element of X which we count, and so on. The function f is an injection because we do not count any element twice and it is a surjection because every element is counted. When we have constructed the bijection $f\colon \mathbb{N}_n \to X$ we can list the elements of the set $X = \{x_1, x_2, \ldots, x_n\}$ where we write $x_i = f(i)$. Thus the function f is described by the following table.

i	1	2	3	$\ldots$	$n-1$	n
$f(i)$	x_1	x_2	x_3	$\ldots$	x_{n-1}	x_n

Because f is a bijection each element of X occurs precisely once in the second row.

We can formulate this as a mathematical definition. It is natural to extend the definition to include the empty set, the set with no elements.†

Definition 10.1.1 *Given a set X, if there is a bijection $f\colon \mathbb{N}_n \to X$ then we say that the* $\boxed{\text{cardinality}}$ *of X, or the* number of elements *in X, is n and write* $\boxed{|X| = n}$.

The cardinality of the empty set is defined to be 0, $\boxed{|\emptyset| = 0}$.

Examples 10.1.2 (a) The cardinality of the set $A = \{\, k \in \mathbb{Z} \mid 25 < k \leqslant 30\}$ is 5. When we list the elements of the set, say $\{29, 26, 30, 27, 28\}$, we are essentially constructing a bijection $f\colon \mathbb{N}_5 \to A$. The bijection corresponding to this listing is given by the following table.

i	1	2	3	4	5
$f(i)$	29	26	30	27	28

This is not the most obvious way of counting this set. The bijection

† In fact we could define $\mathbb{N}_0$ to be the empty set so that the case of cardinality 0 is covered by the bijection definition. However, the reader meeting set theory for the first time may be worried by the bijection between two empty sets (it is the function whose graph is the empty subset of $\emptyset \times \emptyset = \emptyset$!) and so it seems best to keep this case separate.

$g: \mathbb{N}_5 \to A$ corresponding to the obvious way of counting it is given by the following table.

i	1	2	3	4	5
$g(i)$	26	27	28	29	30

(b) The cardinality of the set $\mathbb{N}_n$ is n for the identity map provides a bijection $\mathbb{N}_n \to \mathbb{N}_n$. Another bijection $f: \mathbb{N}_n \to \mathbb{N}_n$ is given by $f(i) = n - i$.

There is one rather subtle difficulty about this definition: we need to be sure that if we count a set in two different ways then we get the same answer. For example, in Example 10.1.2(a), we gave two ways of counting the set $\{ k \in \mathbb{Z} \mid 25 < k \leqslant 30 \}$ and there are others. However, whatever order we use to list this set of integers we find that the fifth element is the last one.

It is common to count a set more than once if we want to be sure about the number of elements in it. If a set is counted twice, for example when counting the number of people in a room, and the answers differ then we assume either that a mistake was made on one or both occasions (in other words the person counting failed to construct a bijection; it can be difficult to be sure that you have an injection and that you have a surjection if people keep moving about!) or that it wasn't the same set (somebody entered or left the room and so was counted on one occasion but not the other). In other words we assume that if mistakes are avoided then the answer will be the same even though we do not assume that the bijections will be the same. We can formulate this assumption as follows.

Proposition 10.1.3 *Suppose that $f: \mathbb{N}_m \to X$ and $g: \mathbb{N}_n \to X$ are bijections with the same codomain. Then $m = n$.*

We need to know this in order that the above definition of the number of elements in a set gives a *well-defined* number. Very often in mathematics definitions are made which involve some choices. Demonstrating that the notion defined is well-defined amounts to proving that it is independent of these choices, in other words different choices lead to the same answer. The notion of cardinality provides a good example of such a definition.

At this point the reader may consider that the author has taken leave of his senses. Surely it is absolutely obvious that the number of elements in a set is a well-defined notion and two different orders of counting cannot lead to a different answer. This may be so, although our experience is only based on relatively small sets and modern physics suggests that care may be needed when we come to deal, say, with a set of electrons. In Chapter 14 we will look briefly at infinite sets and see there how misleading our intuition can be. It is often the case in mathematics that we know what properties we want some notion to have. The real question is how to define the notion mathematically. If we were unable to prove Proposition 10.1.3 then this would call into question whether Definition 10.1.1 is the right one. It has often taken many years to get a definition right, giving an appropriate mathematical formulation of the meaning of a word which successfully captures the intuitive concept. For example the search for the most useful definition of continuity when applied to functions $\mathbb{R} \to \mathbb{R}$ took over a hundred years.

Constructing a proof of Proposition 10.1.3. In spite of being intuitively 'obvious' this is a not an easy result; it would be remarkable for a student to find a proof without some assistance. One difficulty is that it involves a set X about which we know very little. Let us summarize what we are trying to achieve.

Given	**Goal**
bijection $f: \mathbb{N}_m \to X$	
bijection $g: \mathbb{N}_n \to X$	$m = n$

To proceed, notice that the goal does not involve the set X. However, since f and g are bijections they are invertible with inverses $f^{-1}: X \to \mathbb{N}_m$ and $g^{-1}: X \to \mathbb{N}_n$ which are also bijections. So the composite

$$g^{-1} \circ f: \mathbb{N}_m \xrightarrow{f} X \xrightarrow{g^{-1}} \mathbb{N}_n$$

is also a bijection (with inverse $f^{-1} \circ g$). Thus we can use the bijections f and g to construct a bijection $\mathbb{N}_m \to \mathbb{N}_n$. Now the statement that such a bijection exists does not involve the set X. This suggests the following strategy for proving the proposition.

Given	**Goal**
bijection $\mathbb{N}_m \to \mathbb{N}_n$	$m = n$

If we can prove this then we will be home since the given statement follows from the hypotheses in the proposition. Notice that this implication is just a special case of the result we are trying to prove with $X = \mathbb{N}_n$ for we then automatically have a bijection $g: \mathbb{N}_n \to \mathbb{N}_n$ given by the identity map. It is quite common to prove a general result by reducing it to some special case.

Now to tackle this implication we look back to the definition of a bijection: a function is a bijection if and only if it is an injection and a surjection. So we might begin by examining what can be deduced simply from the injectivity of a function $\mathbb{N}_m \to \mathbb{N}_n$. Intuitively this implies that $m \leqslant n$, suggesting the following first step.

Given	Goal
injection $\mathbb{N}_m \to \mathbb{N}_n$	$m \leqslant n$

In fact this implication on its own would be sufficient for our needs. For a bijection $h: \mathbb{N}_m \to \mathbb{N}_n$ is an injection and has an inverse $h^{-1}: \mathbb{N}_n \to \mathbb{N}_m$ which is also a bijection and so an injection. Applying the implication to h gives $m \leqslant n$ and applying it to h^{-1} gives $n \leqslant m$. But $m \leqslant n$ and $n \leqslant m$ only if $m = n$ (see Exercise 4.5).

This process of reducing a result to be proved to successively simpler or more concrete statements is common in constructing proofs. To keep track of what we are doing it can be useful to formulate these other statements as propositions or theorems in their own right; sometimes the word 'lemma' is used instead, indicating that a result's primary interest is as a stepping stone in the proof of some more striking result or results.

In this case it turns out that the implication we have reached is of sufficient interest to be stated formally as a lemma. So we do this now and write out the formal proof that Proposition 10.1.3 follows from it. We will then return to the proof of the lemma in the next chapter.

Proof of Proposition 10.1.3 This result follows from the following result to be proved in the next chapter.

Lemma 10.1.4 *If there exists an injection $\mathbb{N}_m \to \mathbb{N}_n$ then $m \leqslant n$.*

For, given bijections $f: \mathbb{N}_m \to X$ and $g: \mathbb{N}_n \to X$, they are invertible with inverses $f^{-1}: X \to \mathbb{N}_m$ and $g^{-1}: X \to \mathbb{N}_m$ which are also bijections. So the composite $g^{-1} \circ f: \mathbb{N}_m \to \mathbb{N}_n$ is a bijection, and so an injection. It follows from the lemma that $m \leqslant n$.

But now we can reverse the rôles of f and g. The composite $f^{-1} \circ g: \mathbb{N}_n \to \mathbb{N}_m$ is an injection and so again by the lemma $n \leqslant m$.

Since we have proved that $m \leqslant n$ and $n \leqslant m$ it follows that $m = n$ as required. □

Of course not all sets can be counted. For example if we start counting the positive integers in the obvious way we never come to an end. It is useful to distinguish sets for which the counting process does end from those which do not; to distinguish finite sets from infinite sets.

Definition 10.1.5 *Given a set X, if $|X| = n$ for some non-negative integer n then we say that the set is* $\boxed{\text{finite}}$. *If this is not the case then we say that the set is* $\boxed{\text{infinite}}$.

Thus saying that a set X is finite means that either it is empty or there exists a bijection $\mathbb{N}_n \to X$ for some positive integer n.

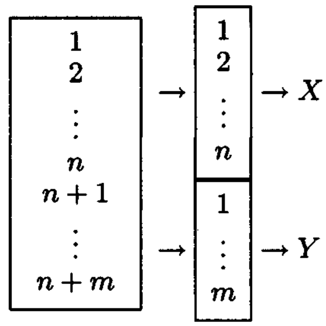
10.2 Two basic counting principles

There are two basic counting principles which are frequently used and extremely powerful. The first seems almost too simple to state.

Theorem 10.2.1 (The addition principle) *Suppose that X and Y are disjoint finite sets. Then $X \cup Y$ is finite and $|X \cup Y| = |X| + |Y|$.*

Constructing a proof. The form of words used in the conclusion of this result sometimes causes confusion. Because it is an 'and' statement the reader might imagine that it is necessary to prove the first statement and then to prove the second statement. But in fact the first statement is an immediate consequence of the second and, with what we have done so far, the only way to prove that a set is finite is to prove that it has cardinality equal to some non-negative integer. Since X and Y are finite sets, $|X|$ and $|Y|$ are non-negative integers and thus so is $|X| + |Y|$. We obtain the result by simply proving that $X \cup Y$ has cardinality equal to $|X| + |Y|$.

Notice that since the set X is finite then either X is empty or there is a bijection $N_n \to X$ for some positive integer n. It is necessary to consider these two cases separately and similarly for Y. The cases when X or Y has cardinality 0 are trivial. In the other cases the result seems obvious since we simply count the elements in the set X, say n, and then count the elements of Y starting at $n + 1$. The following proof simply translates this intuitive idea into the language of bijections, using the bijections to X and Y to give a bijection to $X \cup Y$ as indicated in the diagram on the next page.



Proof Let $|X| = n$ and $|Y| = m$. We are required to prove that $|X \cup Y| = n + m$. If $n = 0$ then X is the empty set and so $X \cup Y = Y$ so that automatically $|X \cup Y| = |Y| = m = 0 + m$. Similarly the result is clear if $m = 0$. So let us consider the case when n and m are both positive.

Since $|X| = n$ and $|Y| = m$ there are bijections $f: \mathbb{N}_n \to X$ and $g: \mathbb{N}_m \to Y$. We can now define a bijection $h: \mathbb{N}_{m+n} \to X \cup Y$ as follows.

$$h(i) = \begin{cases} f(i) & \text{if } 1 \leqslant i \leqslant n, \\ g(i-n) & \text{if } n+1 \leqslant i \leqslant n+m. \end{cases}$$

This shows that $|X \cup Y| = m + n$. Notice that h is an injection because f and g are injections and $X \cap Y = \emptyset$; h is a surjection because f and g are surjections. □

At the end of this proof some details have been omitted. The reader might like to consider how to supply these details justifying the last sentence of the proof. This sentence does indicate how the hypotheses in the theorem are brought into play in the proof; it is important when writing out a proof to indicate at which point(s) hypotheses are used.

Corollary 10.2.2 *For a positive integer n, suppose that $X_1, X_2, \ldots, X_n$ is a collection of n* pairwise disjoint *finite sets (i.e. $i \neq j \Rightarrow X_i \cap X_j = \emptyset$). Then $X_1 \cup X_2 \cup \ldots \cup X_n = \bigcup_{i=1}^{n} X_i$ is a finite set and*

$$|X_1 \cup X_2 \cup \ldots \cup X_n| = |X_1| + |X_2| + \ldots + |X_n|.$$

Proof This is an easy induction proof using the theorem. The details are left as an exercise. □

The second basic counting principle is a consequence of the first.

Theorem 10.2.3 (The multiplication principle) *Suppose that X and Y are finite sets such that $|X| = n$ and $|Y| = m$. Then the Cartesian product $X \times Y$ is a finite set and $|X \times Y| = mn$.*

Proof If either of X and Y is empty then so is $X \times Y$ and the result follows. So suppose that both sets are non-empty.

Let $f: \mathbb{N}_n \to X$ be a bijection and write $x_i = f(i)$. Then

$$X = \{x_1, x_2, \ldots, x_n\} = \{x_1\} \cup \{x_2\} \cup \ldots \cup \{x_n\}$$

and so

$$X \times Y = (\{x_1\} \times Y) \cup (\{x_2\} \times Y) \cup \ldots \cup (\{x_n\} \times Y).$$

But now, if $g: \mathbb{N}_m \to Y$ is a bijection, then $i \mapsto (x_k, g(i))$ gives a bijection $\mathbb{N}_m \to \{x_k\} \times Y$ so that $|\{x_k\} \times Y| = m$. It follows from the previous corollary that $|X \times Y| = mn$. $\square$

Examples 10.2.4 (a) An element of $\mathbb{N}_9 \times \mathbb{N}_9$ is an ordered pair of positive integers each no greater than 9. The number of such pairs is $9 \times 9 = 81$ by the multiplication principle.

(b) What then is the number of ordered pairs of *distinct* positive integers no greater than 9? The function $n \mapsto (n, n)$ provides a bijection $\mathbb{N}_9 \to \{(n, n) \mid n \in \mathbb{N}_9\}$ showing that there are nine ordered pairs of equal integers. This set of ordered pairs is called the *diagonal set* in $\mathbb{N}_9 \times \mathbb{N}_9$ and is denoted by $\Delta(\mathbb{N}_9)$. We are interested in the cardinality of the complement of the diagonal set, $\mathbb{N}_9 \times \mathbb{N}_9 - \Delta(\mathbb{N}_9)$. However, $\mathbb{N}_9 \times \mathbb{N}_9$ is the disjoint union $\Delta(\mathbb{N}_9) \cup (\mathbb{N}_9 \times \mathbb{N}_9 - \Delta(\mathbb{N}_9))$ and so, by the addition principle,

$$|\mathbb{N}_9 \times \mathbb{N}_9| = |\Delta(\mathbb{N}_9)| + |\mathbb{N}_9 \times \mathbb{N}_9 - \Delta(\mathbb{N}_9)|$$

so that the number of ordered pairs of distinct postive integers no greater than 9, $|\mathbb{N}_9 \times \mathbb{N}_9 - \Delta(\mathbb{N}_9)|$, is $81 - 9 = 72$.

There are of course other ways of obtaining the same result. For example, there are nine possibilities for the first coordinate and for each of these there are then eight possibilities for the second coordinate so that the number of ordered distinct pairs is $9 \times 8 = 72$.

From this result we can deduce that the number of *unordered* distinct pairs is $72/2 = 36$ since each pair may be ordered in two ways.

10.3 The inclusion–exclusion principle

The addition principle in the last section leads on to an extremely powerful combinatorial technique known as the inclusion–exclusion principle. This is concerned with a situation where we have a finite number of finite sets of known cardinality (not in general disjoint) and we wish to know the number of elements which lie in their union. Let us consider the simplest example, two sets.

Proposition 10.3.1 *Suppose that X and Y are (not necessarily disjoint) finite sets. Then $X \cup Y$ is a finite set and*

$$|X \cup Y| = |X| + |Y| - |X \cap Y|.$$

Proof Recall from Proposition 6.2.4 that $X \cup Y = (X - Y) \cup (Y - X) \cup (X \cap Y)$, a union of disjoint sets, so that by the addition principle

$$|X \cup Y| = |X - Y| + |Y - X| + |X \cap Y|.$$

Similarly, $X = (X - Y) \cup (X \cap Y)$ is a union of disjoint sets and so

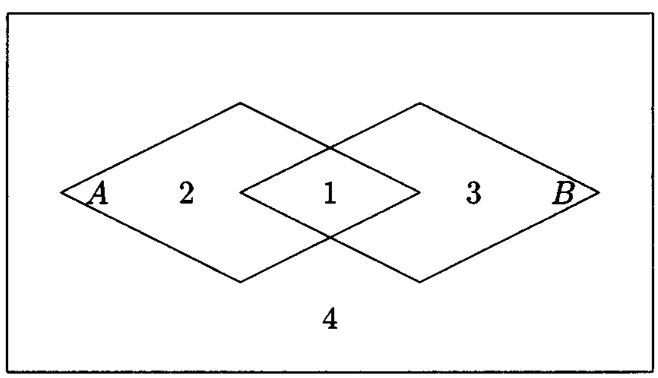
$$|X| = |X - Y| + |X \cap Y|.$$

Similarly $|Y| = |Y - X| + |X \cap Y|$. Thus

$$|X \cup Y| = (|X| - |X \cap Y|) + (|Y| - |X \cap Y|) + |X \cap Y| = |X| + |Y| - |X \cap Y|$$

as required. □

This result is really obvious from the following Venn diagram.



Recall that $A \cup B$ consists of regions 1, 2 and 3; A consists of regions 1 and 2; B consists of 1 and 3; and $A \cap B$ consists of region 1. If we count A and then count B giving $|A| + |B|$ then we will have counted the elements in $A \cap B$ twice and so we take account of this by subtracting $|A \cap B|$.

The inclusion–exclusion principle is the generalization of this result to an arbitrary finite collection of sets. Consider the case of three sets.

Proposition 10.3.2 *Suppose that X, Y and Z are finite sets. Then $X \cup Y \cup Z$ is a finite set and*

$$|X \cup Y \cup Z| = |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z|.$$

Proof We apply Proposition 10.3.1 to $X \cup Y \cup Z = (X \cup Y) \cup Z$ and then apply it again to the sets which arise.

$$\begin{aligned}
|(X \cup Y) \cup Z| &= |X \cup Y| + |Z| - |(X \cup Y) \cap Z|; && (10.1)\\
|X \cup Y| &= |X| + |Y| - |X \cap Y|; && (10.2)\\
|(X \cup Y) \cap Z| &= |(X \cap Z) \cup (Y \cap Z)| &&\\
&= |X \cap Z| + |Y \cap Z| - |(X \cap Z) \cap (Y \cap Z)| &&\\
&= |X \cap Z| + |Y \cap Z| - |X \cap Y \cap Z|. && (10.3)
\end{aligned}$$

Substituting from equations (10.2) and (10.3) into equation (10.1) gives the result. □

The general inclusion–exclusion principle is given as Problems III, Question 4. This proposition illustrates the inductive step in the proof. A number of applications appear later in the book.

Exercises

10.1 Suppose that X is a non-empty finite set of cardinality n. Prove that a set Y has $|Y| = n$ if and only if there exists a bijection $X \to Y$.

10.2 Prove Corollary 10.2.2.

10.3 Let $X = \{a, b, c\}$ and $Y = \{d, e\}$. What is the cardinality of the Cartesian product $X \times Y$? Write down an explicit bijection $\mathbb{N}_n \to X \times Y$ where $n = |X \times Y|$.

10.4 Each of a collection of 144 tiles is either triangular or square, either red or blue, and either wooden or plastic. Given that there are 68 wooden tiles, 69 red tiles, 75 triangular tiles, 36 red wooden tiles, 40 triangular wooden tiles, 38 red triangular tiles, and 23 red wooden triangular tiles, how many blue plastic square tiles are there?

11
Properties of finite sets

There are a number of special techniques available for proving results which involve finite sets. It is clearly possible to determine whether or not a finite set contains an element with a certain property simply by trying each element in turn. More interestingly it is possible on occasion to use 'counting arguments' to demonstrate the existence of an element with a certain property without considering the properties of individual elements at all. We consider these techniques in this chapter.

11.1 The pigeonhole principle

We begin by recalling the key step used in the last chapter to prove that the cardinality of a finite set is a well-defined concept. We left for this chapter the proof of the following result.

Lemma 10.1.4 *If there exists an injection* $\mathbb{N}_m \to \mathbb{N}_n$ *then* $m \leqslant n$.

Constructing a proof. Since this is a statement involving natural numbers a proof by induction is suggested. There are two natural numbers m and n in the statement and so there is a choice: induction on m or induction on n? The following proof is by induction on n. The task of constructing an alternative proof by induction on m is left to the reader (see Problems III, Question 8).

The predicate $P(n)$ in this case is the following universal implication.

$$\forall m \in \mathbb{Z}^+ (\exists \text{ an injection } \mathbb{N}_m \to \mathbb{N}_n \Rightarrow m \leqslant n).$$

So the steps of the inductive proof are as follows.
Base case: It is not difficult to prove by contradiction that if there exists an injection $\mathbb{N}_m \to \mathbb{N}_1$ then $m = 1$. (Note that $m \leqslant 1 \Leftrightarrow m = 1$ for a positive integer m.)

Inductive step: For each $k \geqslant 1$ we must prove an implication which may be summarized as follows.

Given	Goal
$\forall m \in \mathbb{Z}^+ (\exists \text{ injection } \mathbb{N}_m \to \mathbb{N}_k \Rightarrow m \leqslant k)$	$\forall m \in \mathbb{Z}^+ (\exists \text{ injection } \mathbb{N}_m \to \mathbb{N}_{k+1} \Rightarrow m \leqslant k+1)$

The variable m is a 'dummy' variable in both the given statement and the goal. To avoid confusion it is better as usual to replace m by some other symbol, say m_1, in the goal statement so that it becomes $\forall m_1 \in \mathbb{Z}^+ (\exists \text{ injection } \mathbb{N}_{m_1} \to \mathbb{N}_{k+1} \Rightarrow m_1 \leqslant k+1)$. The direct method of proving this is to suppose that $f: \mathbb{N}_{m_1} \to \mathbb{N}_{k+1}$ is an injection and then to seek to deduce that $m_1 \leqslant k+1$. This may be summarized as follows.

Given	Goal
$\forall m \in \mathbb{Z}^+ (\exists \text{ injection } \mathbb{N}_m \to \mathbb{N}_k \Rightarrow m \leqslant k)$	
$m_1 \in \mathbb{Z}^+$	
injection $f: \mathbb{N}_{m_1} \to \mathbb{N}_{k+1}$	$m_1 \leqslant k+1$

It is not possible to apply the given universal implication directly to the injection f since its codomain is $\mathbb{N}_{k+1}$: to apply the given implication we need an injection into $\mathbb{N}_k$ and this may require a change of domain. The details of the proof follow [with additional comments in brackets].

Proof We prove Lemma 10.1.4 by induction on n.
Base case: This statement is trivial for $n = 1$ since, if $f: \mathbb{N}_m \to \mathbb{N}_1$, we must have $f(i) = 1$ for all $i \in \mathbb{N}_m$ and this is not an injection if $m > 1$ since then, for example, $f(1) = f(2) = 1$. Thus if f is an injection then $m = 1 = n$.

[Notice that this is really a proof by contrapositive. What is actually proved is '$m > 1 \Rightarrow f: \mathbb{N}_m \to \mathbb{N}_1$ is not an injection' whose contrapositive is '$f: \mathbb{N}_m \to \mathbb{N}_1$ is an injection $\Rightarrow m \leqslant 1$'.]

Inductive step: Suppose, for some $k \geqslant 1$, that the result is true for $n = k$. We must deduce that it holds for $n = k+1$ so suppose that $f: \mathbb{N}_{m_1} \to \mathbb{N}_{k+1}$ is an injection.

[The aim now is to construct an injection $f_1: \mathbb{N}_m \to \mathbb{N}_k$ for some m to which we can apply the inductive hypothesis.]

We consider two cases.

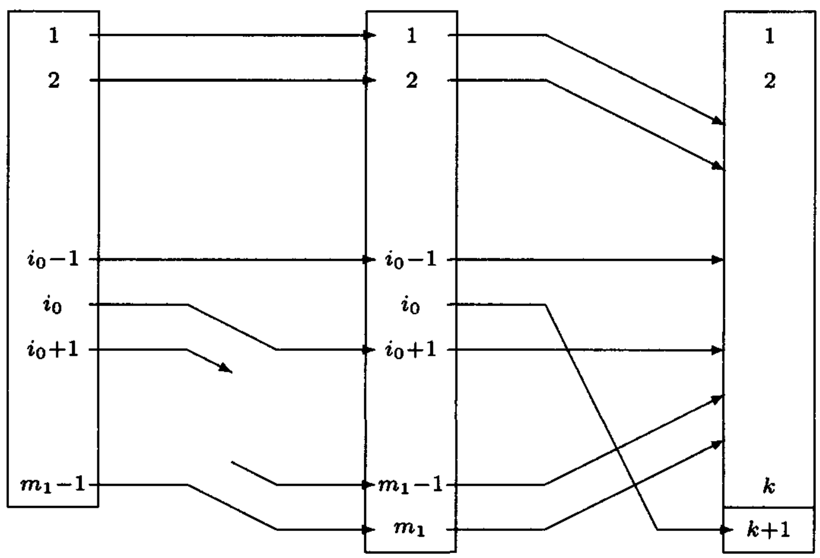
(i) Suppose that $f(i) < k+1$ for all $i \in \mathbb{N}_m$. Then we can restrict the codomain and define a function $f_1: \mathbb{N}_{m_1} \to \mathbb{N}_k$ by $f_1(i) = f(i)$. The function f_1 is also an injection and so, by inductive hypothesis, $m_1 \leqslant k$ and so certainly $m_1 \leqslant k+1$ as required.

[The function f_1 is an injection since $f_1(i_1) = f_1(i_2) \Rightarrow f(i_1) = f(i_2) \Rightarrow i_1 = i_2$ since f is an injection.]

(ii) On the other hand, suppose that $k+1$ is a value of f, say $f(i_0) = k+1$ where $i_0 \in \mathbb{N}_{m_1}$. In this case we can define an injection $g: \mathbb{N}_{m_1-1} \to \mathbb{N}_{m_1}$ by

$$g(i) = \begin{cases} i & \text{for } i < i_0, \\ i+1 & \text{for } i \geqslant i_0, \end{cases}$$

and then define $f_1 = f \circ g: \mathbb{N}_{m_1-1} \to \mathbb{N}_k$.



$$\begin{tikzcd}
\mathbb{N}_{m_1-1} \arrow[rr, "g"] & & \mathbb{N}_{m_1} \arrow[rr, "f"] & & \mathbb{N}_{k+1} \\
\mathbb{N}_{m_1-1} \arrow[u, "\mathrm{Id}"'] \arrow[rrrr, "f_1"] & & & & \mathbb{N}_k \arrow[u, "\text{inclusion}"']
\end{tikzcd}$$

[Notice that if $i \neq i_0$ then $f(i) \neq f(i_0) = k+1$ since f is an injection so that $f(i) \in \mathbb{N}_k$.]

Then this function f_1 is also an injection (since it is the composite of two injections) and so, by inductive hypothesis, $m_1 - 1 \leqslant k$, i.e. $m_1 \leqslant k+1$, as required.

Since one or other of these cases must be true for each function f this completes the proof of the inductive step.

Conclusion: Hence, by induction on n, if $f: \mathbb{N}_m \to \mathbb{N}_n$ is an injection then $m \leqslant n$. $\square$

We can immediately extend this lemma to general finite sets.

Corollary 11.1.1 *Suppose that X and Y are non-empty finite sets. If there exists an injection $f: X \to Y$ then $|X| \leqslant |Y|$.*

So for example if a group of people are all sitting in a room (on separate chairs) then the number of people is no greater than the number of chairs.

Constructing a proof. The hypothesis tells us that there are bijections $g_1: \mathbb{N}_m \to X$ and $g_2: \mathbb{N}_n \to Y$ for some positive integers m and n. Then $|X| = m$ and $|Y| = n$ and we can obtain the result by applying the lemma to the composite

$$g_2^{-1} \circ f \circ g_1: \mathbb{N}_m \to X \to Y \to \mathbb{N}_n$$

since this is an injection (because it is a composite of injections).

This is a good example of how we can deduce a result about general finite sets from the special case about the model sets $\mathbb{N}_n$ of the same cardinality.

Proof By hypothesis $|X| = m$ and $|Y| = n$ for positive integers m and n. Hence there are bijections $g_1: \mathbb{N}_m \to X$ and $g_2: \mathbb{N}_n \to Y$. Then $g_2^{-1} \circ f \circ g_1: \mathbb{N}_m \to \mathbb{N}_n$ is an injection. Hence, by Lemma 10.1.4, $m \leqslant n$, i.e. $|X| \leqslant |Y|$, as required. $\square$

This result is usually stated in the following form. It was first formulated in the first half of the nineteenth century by Lejeune Dirichlet who called it the 'drawer principle'.

Theorem 11.1.2 (The pigeonhole principle) *Suppose that $f: X \to Y$ is a function between non-empty finite sets such that $|X| > |Y|$. Then f is not an injection, i.e. there exist distinct elements x_1 and $x_2 \in X$ such that $f(x_1) = f(x_2)$.*

Proof This is the contrapositive of Corollary 11.1.1 and so follows from that result. $\square$

Examples 11.1.3 (a) There are many simple applications of this result. For example, in any group of more than twelve people there must be two people with birthdays in the same month.

(b) For a slightly more elaborate application consider the following. Suppose that X is a set of at least two people. Then it can be shown that within this set there are two people with the same number of friends in the set. (Here we are assuming that if x_1 is a friend of x_2 then x_2 is a friend of x_1.)

To see this let $|X| = n$ and define a function $f: X \to \{0, 1, 2, \ldots, n-1\}$ by setting $f(x)$ equal to the number of friends of x in X. Now notice that if $n-1$ is a value of f then there exists $x_0 \in X$ such that $f(x_0) = n-1$. This means that the person x_0 is a friend of everyone else and so no one is friendless, i.e. 0 is not a value of f. Hence the numbers 0 and $n-1$ (distinct since $n \geqslant 2$) cannot both be values of f. Hence the image $\mathrm{Im} f$ has cardinality less than n and so by the pigeonhole principle f cannot be an injection, i.e. there exist distinct elements x_1 and x_2 such that $f(x_1) = f(x_2)$ as claimed.

Some more mathematical applications will come later.

We can mimic the proof of Lemma 10.1.4 to prove the following more general result. The details are left to the reader (see Problems III, Question 9). (A similar result appears as Exercise 11.1.)

Proposition 11.1.4 *Suppose that $f: X \to \mathbb{N}_n$ is an injection. Then X is a finite set and $|X| \leqslant n$.*

This has the following consequence which is intuitively completely obvious. However, a *proof* of it has to be based on our definitions.

Corollary 11.1.5 *Suppose that $X \subseteq Y$ where Y is a finite set. Then X is also a finite set and $|X| \leqslant |Y|$.*

Proof Suppose that $|Y| = n$ so that there is a bijection $f: \mathbb{N}_n \to Y$. Let $i: X \to Y$ be the inclusion function $i(x) = x$. This is clearly an injection and so the composite $f^{-1} \circ i: X \to Y \to \mathbb{N}_n$ is an injection. Hence, by Proposition 11.1.4, X is finite and $|X| \leqslant n$, as required. □

In the above we have concentrated on the property of injectivity. It is not difficult to deduce similar results about surjectivity such as the following.

Theorem 11.1.6 *Suppose that $f: X \to Y$ is a function between non-empty finite sets such that $|X| < |Y|$. Then f is not a surjection, i.e. there exists an element of Y which is not a value of the function.*

This can be proved by similar methods to the pigeonhole principle. Alternatively it can be deduced from the pigeonhole principle by observing that from a surjection $X \to Y$ it is possible to construct an injection $Y \to X$ (see for example Problems II, Question 19).

If two finite sets are known to have the same size, then it becomes easier to check whether a function between them is a bijection.

Theorem 11.1.7 *Suppose that X and Y are non-empty finite sets of the same cardinality. Then a function $X \to Y$ is an injection if and only if it is a surjection. Thus to prove bijectivity it is only necessary to check one of injectivity and surjectivity.*

The proof of this result is left to the reader (Problems III, Question 11).

11.2 Finite sets of real numbers

One advantage of working with finite sets is that when we are seeking an object with a certain property each of the elements can be tested in turn and this task will terminate. As an example of this consider the problem of seeking the greatest element in a set of real numbers. Of course there are sets of numbers without a greatest, for example the set of integers (see Problems I, Question 10). It will now be shown that any *finite* non-empty set of real numbers does have a greatest element.

Definition 11.2.1 *Let A be a set of real numbers, i.e. $A \subseteq \mathbb{R}$. Then b is a* $\boxed{\text{minimum element of } A}$ *or* $\boxed{b = \min A}$ *when*

(i) *b is an element of A, i.e. $b \in A$,*
(ii) *b is less than or equal to every element of A, i.e. $a \in A \Rightarrow b \leqslant a$.*

Similarly, c is a $\boxed{\text{maximum element of } A}$ *or* $\boxed{c = \max A}$ *when*

(i) *c is an element of A, i.e. $c \in A$,*
(ii) *c is greater than or equal to every element of A, i.e. $a \in A \Rightarrow c \geqslant a$.*

Notice that if a set of real numbers, A, has a maximum then this maximum is unique so that the number $\max A$ is well-defined. Writing out a formal proof of this uniqueness is left as an exercise (Exercise 11.2). Similarly the minimum is unique if it exists.

Examples 11.2.2 (a) For $A = \{-1, 17, 12, -78, 8\}$, $\min A = -78$ and $\max A = 17$.

(b) The set $\mathbb{Z}^+$ of positive integers has $\min \mathbb{Z}^+ = 1$ but there is no maximum element.

(c) Any non-empty set of positive integers has a minimum element. This result is known as *the well-ordering principle* and is equivalent to the induction principle (see Exercise 11.6 and Problems III, Question 15).

(d) The set $\mathbb{R}^+ = \{x \in \mathbb{R} \mid x > 0\}$ of positive real numbers has no minimum element and no maximum element. Notice that, given any element $a \in \mathbb{R}^+$ there is a smaller one, for example $a/2 \in \mathbb{R}^+$ (see Problems I, Question 11).

Proposition 11.2.3 *Let A be a finite non-empty set of real numbers. Then A has a minimum element and a maximum element.*

When this result is applied it is always important to check that the set under consideration is non-empty. The empty set is finite but does not have a minimum or maximum element since it has no elements at all!

Constructing a proof. If we think why this is intuitively obvious we come up with the following ideas.

Suppose that $|A| = n \in \mathbb{Z}^+$. (This makes use of the fact that A is finite.) Then we can write $A = \{a_1, a_2, \ldots, a_n\}$. To find the maximum element we try a_1 and begin successively comparing it with the other elements a_2, a_3 and so on. If we hit something greater we start using that in place of a_1 and carry on working through the list repeating the replacement process whenever we reach a greater element. When we have completed the list, the element we have is the maximum element.

To write out this argument more formally we notice that the phrase 'and so on' suggests that this is really an induction argument even though the statement of the proposition does not explicitly refer to a positive integer. However, there is an implicit integer, namely the cardinality of the set A.

Proof We write out the proof for 'maximum element' leaving the proof for 'minimum element' as an exercise.

We prove by induction on n that, for all positive integers n, all non-empty finite sets of real numbers with cardinality n have a maximum element. This is clearly equivalent to the proposition because any non-empty finite set has cardinality n for some positive integer n, by definition of 'finite'.

Base case: For $n = 1$, the single element of the set is the maximum (and the minimum) element!

Inductive step: Suppose that for some $k \in \mathbb{Z}^+$ the result holds for $n = k$ so that all sets of real numbers of cardinality k have a maximum element. Let $A = \{a_1, a_2, \ldots, a_k, a_{k+1}\}$ be a set of real numbers of cardinality $k + 1$. By inductive hypothesis, the set $A' = \{a_1, a_2, \ldots, a_k\}$ has a maximum element, say a'. If $a' \geqslant a_{k+1}$, then a' is the maximum element of A. If $a' < a_{k+1}$, then a_{k+1} is the maximum element of A. In either case A has a maximum element and so we have deduced the result for $n = k + 1$, completing the inductive step.

Hence by induction the result holds for all $n \in \mathbb{Z}^+$ and the proposition is proved. □

11.3 Two applications of finiteness

The greatest common divisor

In Definition 2.2.1 we introduced the notion of one integer dividing another. Recall that a non-zero integer d divides a when there is an integer q such that $a = dq$. In this case we say that d is a *divisor* or a *factor* of a. Alternatively we say that a is a *multiple* of d.

Every non-zero integer divides 0 (since we can take $q = 0$). However, if a is non-zero then, if $a = dq$, q is also non-zero and so, since $|q| \geqslant 1$, we have $|a| \geqslant |d|$. Thus the set of divisors of a,

$$D(a) = \{\, n \in \mathbb{Z} \mid n \text{ divides } a \,\},$$

is finite if a is non-zero with $\min D(a) = -|a|$ and $\max D(a) = |a|$.

If now we are given two integers a and b, not both zero, then the set of *common divisors* of a and b is given by the intersection $D(a) \cap D(b)$. Notice that $1 \in D(a) \cap D(b)$ so that $D(a) \cap D(b)$ is a non-empty finite set and so must then have a maximum element: the *greatest common divisor.*

Definition 11.3.1 *Let a and b be two integers, at least one of which* *is non-zero. Then the* $\boxed{\text{greatest common divisor}}$ *or* highest common factor *of* a *and* b *is the unique positive integer* d *such that*

(i) d *is a common divisor, i.e.* d *divides* a *and* d *divides* b,
(ii) d *is greater than every other common divisor:*

$$c \text{ divides } a \text{ and } c \text{ divides } b \Rightarrow c \leqslant d.$$

We denote the greatest common divisor of a *and* b *by* $\boxed{(a,b)}$† *or in case of possible ambiguity* $\boxed{\gcd(a,b)}$.

Definition 11.3.2 *Two integers* a *and* b*, not both zero, are called* $\boxed{\text{coprime}}$ *or* relatively prime *when* $(a,b) = 1$*, in other words their only common factors are* 1 *and* -1.

Example 11.3.3 The set of divisors of 30 is

$$D(30) = \{-30, -15, -10, -6, -5, -3, -2, -1, 1, 2, 3, 5, 6, 10, 15, 30\}$$

and the set of divisors of 72 is

$$\begin{aligned} D(72) &= \{-72, -36, -24, -18, -12, -9, -8, -6, -4, \\ &\qquad -3, -2, -1, 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72\}. \end{aligned}$$

Hence the set of common divisors is

$$D(30, 72) = \{-6, -3, -2, -1, 1, 2, 3, 6\}.$$

Thus the greatest common divisor $(30, 72) = 6$.

Odd and even integers

We can use these ideas to prove a result which was used in Problems I, Question 7. It is a special case of the *division theorem* which will be reached in Chapter 15.

† The reader should be aware (and beware) of the way in which some notation is used in a variety of ways. Here we are using (a, b) for the greatest common divisor of two integers, but we have previously used this notation to devote an element of the Cartesian product $\mathbb{Z} \times \mathbb{Z}$ and to denote the open interval $\{x \in \mathbb{R} \mid a < x < b\}$. It is usually clear from the context which of these meanings is intended. The reader just has to get used to this sort of ambiguity.

Proposition 11.3.4 *An integer a is odd if and only if $a = 2q + 1$ for some integer q.*

Recall that we have defined 'odd' to mean not divisible by 2 (see Definition 2.2.3). Since this result is an 'if and only if' statement we have two things to prove: that the condition $a = 2q + 1$ for some q is both necessary and sufficient for a to be odd. The proof of sufficiency is a generalization of the proof of Proposition 2.2.4 in Section 4.1 so let us get that out of the way.

Proof of the sufficiency of $a = 2q + 1$ Suppose that $a = 2q + 1$ where $q \in \mathbb{Z}$. To prove that a is odd, suppose for contradiction that a is even. Then $a = 2p$ for some $p \in \mathbb{Z}$. But then $2p = a = 2q + 1$ so that $1 = 2(p - q)$ which implies $1 \geqslant 2$ giving the required contradiction.

Hence $2q + 1$ is not even and so is odd by definition. □

Constructing a proof of the necessity of $a = 2q+1$. The difficulty of the proof of necessity is that we somehow have to get our hands on the element q. One common way of constructing an element in results of this type is to define it in terms of some set – in this case we define it as the maximum member of a finite set. If you think of how you would in practice find this number q you realize that you would keep doubling integers until you got as close to a as possible. When we formalize this into a mathematical argument it gives the proof. It is easier to work with positive integers first and then extend to negative ones.

Proof of the necessity of $a = 2q + 1$ (a) Suppose that a is a positive odd integer. Set

$$A = \{\, k \in \mathbb{Z} \mid k \geqslant 0 \text{ and } a \geqslant 2k \,\}.$$

This is a non-empty set since $0 \in A$ and it is a finite set since $k \in A \Rightarrow 2k \leqslant a \Rightarrow k \leqslant a$. So we can define $q = \max A$.

To prove that $a = 2q + 1$ notice first of all that, since $q \in A$, $a \geqslant 2q$ and, since q is odd, $a \neq 2q$ and so $a \geqslant 2q + 1$. On the other hand, since $q = \max A$, $q+1 \notin A$ so that $a < 2(q+1) = 2q+2$, i.e. $a \leqslant 2q+1$. Thus we have proved that $a \geqslant 2q + 1$ and $a \leqslant 2q + 1$ from which it follows that $a = 2q + 1$ as required.

(b) To complete the proof suppose that a is a negative odd integer. Then $-a$ is a positive odd integer so that from what we have proved $-a = 2q_1 + 1$. But then $a = -2q_1 - 1 = 2(-q_1 - 1) + 1 = 2q + 1$ as required if we put $q = -q_1 - 1$. □

Exercises

11.1 Suppose that $\mathbb{N}_n \to X$ is a surjection. Prove, by induction on n, that X is a finite set and that $|X| \leqslant n$.
[This is similar to Proposition 11.1.4.]

11.2 Use Definition 11.2.1 to prove that if a set of real numbers has a maximum element then this element is unique.
[Hint: Suppose that c_1 and c_2 are two maximum elements of a set A and use the definition to prove that $c_1 = c_2$.]

11.3 Find the greatest common divisor of 88 and 136.

11.4 Prove that, if a and b are non-zero integers with $\gcd(a, b) = d$, then the integers a/d and b/d are coprime.

11.5 Suppose that X and Y are finite sets such that $X \subseteq Y$. Prove that

(i) $X = Y \Leftrightarrow |X| = |Y|$,
(ii) $X \subset Y \Leftrightarrow |X| < |Y|$.

11.6 Prove by induction on n that if A is a set of positive integers without a least element then $\mathbb{N}_n \subseteq \mathbb{Z}^+ - A$ for every n so that A is the empty set.

Deduce the well-ordering principle: every non-empty set of positive integers has a least element.

12
Counting functions and subsets

Very many counting problems can be formulated in terms of counting the number of functions between two sets (possibly satisfying certain properties) or counting the number of subsets of a given set (possibly satisfying certain properties). In this chapter we give a brief introduction to these ideas. They naturally lead to the *binomial coefficients*, one of the most important families of numbers in all mathematics.

12.1 Counting sets of functions

Suppose that X and Y are finite sets. It is natural to ask how many different functions there are with domain X and codomain Y. For example, if X is a set of people and Y is a set of dishes on a menu, then each function $X \to Y$ represents a choice of (one) dish for each person, so that the number of functions represents the number of possible orders by the set of people. If S is a set of students and T is a set of tutors then a function $S \to T$ is an assignment of a tutor for each student.

There is another increased level of abstraction here. The idea of treating a set, for example the set of integers, as a single mathematical object is one stage of abstraction. A further stage is viewing a function from one set to another as a single mathematical object. But now we are forming a new set whose elements are the functions from one set to another and asking questions about this set.

Let us return to the problem we are considering to give an example.

Example 12.1.1 Let $X = \{a, b, c\}$ and $Y = \{d, e\}$. Then the eight functions $X \to Y$ are listed in Example 8.1.3. In order to define a function $f: X \to Y$, we must list the elements $f(a)$, $f(b)$ and $f(c)$. Intuitively, since $f(a)$ can be chosen in two ways (d and e), $f(b)$ can be chosen in two ways and $f(c)$ can be chosen in two ways, f can be chosen in $2 \times 2 \times 2 = 8$ ways and so there are eight functions. This example illustrates the following general result.

Proposition 12.1.2 *Suppose that X and Y are non-empty finite sets with $|X| = m$ and $|Y| = n$. Then the number of functions $X \to Y$ is given by n^m.*

Constructing a proof. We can generalize the above argument as follows. By the hypotheses in the proposition we can list the elements of the sets: $X = \{x_1, x_2, \ldots, x_m\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$. A function $f: X \to Y$ is determined by listing the values $f(x_1)$, $f(x_2)$, and so on up to $f(x_m)$. There are n possibilities for $f(x_1)$, n possibilities for $f(x_2)$, and so on. Hence the number of functions is $n \times n \times \ldots \times n$ (m factors) $= n^m$, as required.

The phrase 'and so on' in this argument indicates that it is really an argument by induction – in this case an induction on m. This is a good example of a result with a convincing informal proof, as given above, for which the formal proof is rather complicated. The reader should endeavour to see how the simple idea in the informal proof underpins the formal proof. When writing out a formal proof it is convenient to have a name for the set of functions $X \to Y$.

Definition 12.1.3 *Given sets X and Y we denote the set of functions from X to Y by* $\boxed{\mathrm{Fun}(X, Y)}$.

Proof of Proposition 12.1.2 The proof is by induction on the positive integer m.

Base case: For $m = 1$ suppose that $X = \{x_1\}$. Then for any set Y there is a bijection $\mathrm{Fun}(X, Y) \to Y$ given by $f \mapsto f(x_1)$. Hence $|\mathrm{Fun}(X, Y)| = |Y| = n$ as required.

Inductive step: Suppose that, for some $k \geqslant 1$, the result holds for $m = k$. To deduce the result for $m = k+1$ suppose that X_1 and Y are non-empty finite sets with $|X_1| = k + 1$ and $|Y| = n$. Let $X_1 = \{x_1, x_2, \ldots, x_{k+1}\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$. Then there are n possibilities for $f(x_{k+1})$ and

$$\mathrm{Fun}(X_1, Y) = \bigcup_{i=1}^{n} \{\, f: X_1 \to Y \mid f(x_{k+1}) = y_i \,\},$$

a disjoint union. We can determine the cardinality of each of the sets in the union using the bijection

$$\{ f: X_1 \to Y \mid f(x_{k+1}) = y_i \} \to \mathrm{Fun}(\{x_1, x_2, \ldots, x_k\}, Y)$$

defined by $f \mapsto f|(X_1 - \{x_{k+1}\})$. [Notice that if we know that $f(x_{k+1}) = y_i$ then f is determined by its values on $x_1, \ldots, x_k$.] For then, by inductive hypothesis,

$$|\{ f: X_1 \to Y \mid f(x_{k+1}) = y_i \}| = |\mathrm{Fun}(\{x_1, x_2, \ldots, x_k\}, Y)| = n^k.$$

Hence, by the addition principle,

$$|\mathrm{Fun}(X_1, Y)| = \sum_{i=1}^{n} n^k = n \times n^k = n^{k+1}$$

as required to deduce the result for $m = k + 1$ and so completing the inductive step.
Conclusion: Hence, by induction on m, the result holds for all positive integers m. □

For certain applications we do not wish to know the total number of functions, but just the number which have some specific property. Quite commonly we want to know the number of injections.

Proposition 12.1.4 *Suppose that X and Y are non-empty sets with $|X| = m$ and $|Y| = n$. Then the number of injections $X \to Y$ is given by $n(n-1)\ldots(n-m+1)$.*

The number $n(n-1)\ldots(n-m+1)$ is usually called a *falling factorial* and is denoted by $(n)_m$ in books on combinatorics. When $m = n$ it is just the usual factorial denoted by $n!$. When $m > n$ the falling factorial vanishes since one of the factors is 0: this shows that Proposition 12.1.4 is a generalization of the pigeonhole principle which asserts that in this case there are no injections. If $m \leqslant n$, then $(n)_m = n!/(n-m)!$.

The proof of this result is almost identical to that of Proposition 12.1.2 except that as we choose the values $f(x_1)$, $f(x_2)$ and so on we must avoid elements of the codomain Y which have already been selected. To write it out it is again useful to have a name for the set we are counting.

Definition 12.1.5 *Given sets X and Y we denote the set of injections from X to Y by* $\boxed{\mathrm{Inj}(X, Y)}$.

Proof of Proposition 12.1.4 The proof is by induction on the positive integer m.
Base case: For $m = 1$ suppose that $X = \{x_1\}$. Then for any set Y there is a bijection $\mathrm{Inj}(X,Y) = \mathrm{Fun}(X,Y) \to Y$ given by $f \mapsto f(x_1)$. Hence $|\mathrm{Inj}(X,Y)| = |Y| = n$ as required.
Inductive step: Suppose that, for some $k \geqslant 1$, the result holds for $m = k$. To deduce the result for $m = k+1$ suppose that X_1 is a set of $k+1$ elements. Let $X_1 = \{x_1, x_2, \ldots, x_{k+1}\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$. Then

$$\mathrm{Inj}(X_1,Y) = \bigcup_{i=1}^{n} \{\, f \in \mathrm{Inj}(X_1,Y) \mid f(x_{k+1}) = y_i \,\},$$

a disjoint union. We can determine the cardinality of each of the sets in the union using the bijection

$$\{\, f \in \mathrm{Inj}(X_1,Y) \mid f(x_{k+1}) = y_i \,\} \to \mathrm{Inj}(\{x_1, x_2, \ldots, x_k\}, Y - \{y_i\})$$

defined by $f \mapsto f|(X_1 - \{x_{k+1}\})$. [Notice that if f is an injection and $f(x_{k+1}) = y_i$ then none of the values $f(x_1)$, $\ldots$, $f(x_k)$ can be y_i.] For then, by inductive hypothesis,

$$\begin{aligned} |\{\, f \in \mathrm{Inj}(X_1,Y) \mid f(x_{k+1}) = y_i \,\}| &= |\mathrm{Inj}(\{x_1, x_2, \ldots, x_k\}, Y - \{y_i\})| \\ &= (n-1)_k \\ &= (n-1)(n-2)\ldots(n-k). \end{aligned}$$

Hence, by the addition principle,

$$\begin{aligned} |\mathrm{Inj}(X_1,Y)| &= \sum_{i=1}^{n} (n-1)(n-2)\ldots(n-k) \\ &= n \times (n-1)\ldots(n-k) = (n)_{k+1} \end{aligned}$$

as required to deduce the result for $m = k+1$ and so completing the inductive step.
Conclusion: Hence, by induction on m, the result holds for all positive integers m. □

In the case when $m = n$, an injection is necessarily a bijection (see Theorem 11.1.7). Bijections with the same domain and codomain are of particular interest.

Definition 12.1.6 *Given a set X, a bijection $X \to X$ is called a* $\boxed{\text{permutation}}$ *of the set X.*

Corollary 12.1.7 *Given a finite non-empty set X of cardinality n, the number of permutations of X is given by the factorial $n!$.*

Proof This is just the special case of Proposition 12.1.4 when $m = n$ for in this case an injection is automatically a bijection (by Theorem 11.1.7). □

12.2 Counting sets of subsets

Recall from Definition 6.3.1 that the *power set* of X is the set of subsets of X, i.e.

$$\mathcal{P}(X) = \{ A \mid A \subseteq X \}.$$

Proposition 12.2.1 *Suppose that X is a set of cardinality n. Then the power set $\mathcal{P}(X)$ is a finite set of cardinality 2^n:*

$$|\mathcal{P}(X)| = 2^{|X|}.$$

Constructing a proof. A subset $A \subseteq X$ is determined by deciding whether each of the n elements of X is in the subset: there are clearly two possibilities for each element x, namely $x \in A$ and $x \notin A$. Hence the total number of possibilities for the subset A is given by $2 \times 2 \times \ldots \times 2 = 2^n$.

This rather informal argument can be made precise by using the idea of the *characteristic function* of a subset which appeared in Problems II, Question 15.

Definition 12.2.2 *For any set X, we can associate to each subset A of X ($A \in \mathcal{P}(X)$) a* $\boxed{\text{characteristic function}}$ $\boxed{\chi_A \colon X \to \{0, 1\}}$ *defined by*

$$\chi_A(x) = \begin{cases} 0 & \textit{if } x \notin A, \\ 1 & \textit{if } x \in A. \end{cases}$$

Lemma 12.2.3 *The function $\mathcal{P}(X) \to \mathrm{Fun}(X, \{0, 1\})$ given by $A \mapsto \chi_A$ is a bijection.*

Proof The inverse is given by $\chi \mapsto \{ x \in X \mid \chi(x) = 1 \}$ $\left(= \overleftarrow{\chi}(\{1\})\right)$. □

Proof of Proposition 12.2.1 Using the result on counting functions in the last section,

$$|\mathcal{P}(X)| = |\mathrm{Fun}(X, \{0,1\})| = 2^{|X|},$$

as required. □

Definition 12.2.4 *Given a set X and a non-negative integer r, an* $\boxed{r\text{-subset}}$ *of X is a subset $A \subseteq X$ of cardinality r. We denote the set of r-subsets of X by $\mathcal{P}_r(X)$, i.e.*

$$\mathcal{P}_r(X) = \{ A \subseteq X \mid |A| = r \}.$$

We define the $\boxed{\text{binomial coefficient}}$ *or* binomial number $\boxed{\binom{n}{r}}$ *(read 'n choose r') to be the cardinality of the set $\mathcal{P}_r(X)$ when $|X| = n$.*

The reason for this name will be apparent when we come to the binomial theorem later in this chapter. Notice that given two finite sets X_1 and X_2 of the same cardinality, there is a bijection $f: X_1 \to X_2$. Such a bijection will induce a bijection $\overrightarrow{f}: \mathcal{P}_r(X_1) \to \mathcal{P}_r(X_2)$ (see Definition 9.3.1) and so the binomial coefficients are well-defined.

Example 12.2.5 Suppose that $X = \{ a, b, c, d \}$; then

$$\begin{aligned}
\mathcal{P}_0(X) &= \{\emptyset\}, \\
\mathcal{P}_1(X) &= \{\{a\}, \{b\}, \{c\}, \{d\}\}, \\
\mathcal{P}_2(X) &= \{\{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{b,d\}, \{c,d\}\}, \\
\mathcal{P}_3(X) &= \{\{a,b,c\}, \{a,b,d\}, \{a,c,d\}, \{b,c,d\}\}, \\
\mathcal{P}_4(X) &= \{X\}, \\
\mathcal{P}_r(X) &= \emptyset \quad \text{for } r > 4.
\end{aligned}$$

Thus $\binom{4}{0} = 1$, $\binom{4}{1} = 4$, $\binom{4}{2} = 6$, $\binom{4}{3} = 4$, $\binom{4}{4} = 1$ and $\binom{4}{r} = 0$ for $r > 4$.

This example illustrates the following general results.

Proposition 12.2.6 *For n and r non-negative integers*

(i) $\binom{n}{r} = 0$ *if $r > n$,*

(ii) $\binom{n}{0} = 1$, $\binom{n}{1} = n$, $\binom{n}{n} = 1$,

(iii) $\dbinom{n}{r} = \dbinom{n}{n-r}$ *for* $0 \leqslant r \leqslant n$.

Proof Suppose that X is a finite set with $|X| = n$.

(i) X has no subsets of cardinality greater than n and so $\binom{n}{r} = 0$ for $r > n$.

(ii) $\mathcal{P}_0(X) = \{\emptyset\}$ and so $\binom{n}{0} = 1$. $\mathcal{P}_n(X) = \{X\}$ and so $\binom{n}{n} = 1$.

If $X = \{x_1, x_2, \ldots, x_n\}$ then $\mathcal{P}_1(X) = \big\{\{x_1\}, \{x_2\}, \ldots, \{x_n\}\big\}$ and so $\binom{n}{1} = n$.

(iii) There is a bijection $\mathcal{P}_r(X) \to \mathcal{P}_{n-r}(X)$ given by mapping each subset to its complement $A \mapsto A^c$. Hence $\binom{n}{r} = \binom{n}{n-r}$. □

Proposition 12.2.7

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n.$$

Proof This is immediate from Definition 12.2.4 using Proposition 12.2.1 and Proposition 12.2.6(i). □

Evaluating binomial coefficients

The method used in Example 12.2.5, calculating binomial coefficients by enumerating subsets, is extremely inefficient. There is an ancient inductive method for evaluating these numbers and the key to this is the following result.

Proposition 12.2.8 *For positive integers n and r such that* $1 \leqslant r \leqslant n$

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}.$$

Proof Let X be a set of cardinality n and choose $x_1 \in X$. Then we may define a bijection $f: \mathcal{P}_r(X) \to \mathcal{P}_{r-1}(X - \{x_1\}) \cup \mathcal{P}_r(X - \{x_1\})$ as follows. Let $A \in \mathcal{P}_r(X)$. Then

$$f(A) = \begin{cases} A - \{x_1\} \in \mathcal{P}_{r-1}(X - \{x_1\}) & \text{if } x_1 \in A, \\ A \in \mathcal{P}_r(X - \{x_1\}) & \text{if } x_1 \notin A. \end{cases}$$

We can prove that this is a bijection by writing down the inverse

$$g: \mathcal{P}_{r-1}(X - \{x_1\}) \cup \mathcal{P}_r(X - \{x_1\}) \to \mathcal{P}_r(X)$$

as follows:

$$g(A) = \begin{cases} A \cup \{x_1\} & \text{if } A \in \mathcal{P}_{r-1}(X - \{x_1\}), \\ A & \text{if } A \in \mathcal{P}_r(X - \{x_1\}). \end{cases}$$

The result now follows by the addition principle (Theorem 10.2.1). $\square$

Remarks 12.2.9 This relation gives an inductive method for calculating binomial coefficients which appears to have been first discovered in China by the end of the eleventh century but is now known in the west as Pascal's triangle after the French mathematician Blaise Pascal (1623–1662) who connected the study of probability with the binomial coefficients and made use of this method of calculating them.

We may display the binomial coefficients in a triangle beginning as follows.

$$\begin{array}{ccccccccccccccc}
 & & & & & & & 1 & & & & & & & \\
 & & & & & & 1 & & 1 & & & & & & \\
 & & & & & 1 & & 2 & & 1 & & & & & \\
 & & & & 1 & & 3 & & 3 & & 1 & & & & \\
 & & & 1 & & 4 & & 6 & & 4 & & 1 & & & \\
 & & 1 & & 5 & & 10 & & 10 & & 5 & & 1 & & \\
 & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 & \\
1 & & 7 & & 21 & & 35 & & 35 & & 21 & & 7 & & 1
\end{array}$$

Here the numbers in the $(n+1)$st row are the binomial coefficients $\binom{n}{r}$ for $r = 0, 1, 2, \dots, n$. By Proposition 12.2.6 the border consists entirely of the number 1. By Proposition 12.2.8 each number is the sum of the two adjacent numbers in the row above. This is not a very convenient method of working out a single binomial coefficient but there is an explicit formula as follows.

Theorem 12.2.10 *For non-negative integers n and r such that $r \leqslant n$,*

$$\binom{n}{r} = \frac{n(n-1)\dots(n-r+1)}{r!} = \frac{n!}{r!(n-r)!}.$$

Constructing a proof. The most direct proof of this formula is by induction using Proposition 12.2.8. The details of this argument are given below. However, although this is probably the simplest argument to write out clearly it is not the most enlightening argument: it gives no indication why the result is true. Mathematicians prefer proofs which give some understanding although this is not always achieved.†

In this case we can find an alternative and, I would argue, a better argument based on the definition of the binomial number as the cardinality of a set of subsets of a given cardinality and intuitive ideas about how we choose such a subset.

Suppose that X is a set of n elements. Each element A of $\mathcal{P}_r(X)$ may be specified by listing the r distinct elements of X which belong to A. For $r > 0$ this amounts to constructing an injection $f\colon \mathbb{N}_r \to X$ so that $\operatorname{Im}(f) = A$. Thus the function

$$\phi\colon \operatorname{Inj}(\mathbb{N}_r, X) \to \mathcal{P}_r(X)$$

given by $\phi(f) = \operatorname{Im}(f)$ is a surjection. However, it is not a bijection since different injections may have the same image: we may list the elements of a subset A in a variety of orders. Each ordering of the elements of A corresponds to an element of the set $\operatorname{Inj}(\mathbb{N}_r, A)$ and, by Proposition 12.1.4, this set has cardinality $r!$. Hence, since the cardinality of $\operatorname{Inj}(\mathbb{N}_r, X)$ is $n!/(n-r)!$ (also by Proposition 12.1.4), it follows that the cardinality of $\mathcal{P}_r(X)$ is $(n!/(n-r)!)/r! = n!/r!(n-r)!$ as claimed in the theorem.

Proof The proof is by induction on n.
Base case: If $n = 0$ then the only possibility for r is $r = 0$ and we know that $\binom{0}{0} = 1$ from Proposition 12.2.6. When $n = 0$ and $r = 0$, $n!/r!(n-r)! = 0!/0!0! = 1$ (recall that $0! = 1$) and so the result is true for $n = 0$.
Inductive step: Suppose as inductive hypothesis that, for some $k \geqslant 0$, the result is true for $n = k$. To deduce it for $n = k+1$, first observe that, for $r = 0$ and $r = k+1$, $\binom{k+1}{r} = 1$ (by Proposition 12.2.6) and $(k+1)!/r!(k+1-r)! = (k+1)!/0!(k+1)! = 1$ and so the result holds in these cases. Now suppose that $1 \leqslant r \leqslant k$. Then, by Proposition 12.2.8,

$$\binom{k+1}{r} = \binom{k}{r-1} + \binom{k}{r}$$

† The eminent mathematician Pierre Deligne once wrote at the end of a very formal proof of his own 'I would be grateful if anyone who has understood this proof would explain it to me', '*Je serais reconnaissant à toute personne ayant compris cette démonstration de me l'expliquer*' (see *Théorie des topos et cohomologie étale des schémas, Tome 3,* Lecture Notes in Mathematics, **305**, Springer-Verlag, 1973, page 584). This is quoted in D. Alibert and M. Thomas, 'Research into mathematical proof' in David Tall (editor), *Advanced mathematical thinking,* Kluwer, 1991.

$$= \frac{k!}{(r-1)!(k-r+1)!} + \frac{k!}{r!(k-r)!}$$
by inductive hypothesis
$$= \frac{k!r}{r!(k-r+1)!} + \frac{k!(k-r+1)}{r!(k-r+1)!}$$
$$= \frac{k!(r+k-r+1)}{r!(k-r+1)!} = \frac{(k+1)!}{r!(k+1-r)!}$$

as required to deduce the result for $n = k+1$. This proves the inductive step.
Hence the result is true for all positive integers n. □

12.3 The binomial theorem

[Professor Moriarty] is ... endowed by nature with a phenomenal mathematical faculty. At the age of twenty-one he wrote a treatise upon the binomial theorem, which has had a European vogue. On the strength of it he won the mathematical chair at one of our smaller universities, and had, to all appearances, a most brilliant career before him.

Arthur Conan Doyle, *Memoirs of Sherlock Holmes* (The final problem).

The name 'binomial coefficients' for the family of numbers considered in the previous section comes from its rôle in the binomial theorem. This theorem describes how to remove the brackets from an expression of the form $(a+b)^n$. A *binomial* is the sum of two terms and so $a+b$ is a binomial.

Theorem 12.3.1 (The binomial theorem) *For all real numbers a and b and non-negative integers n,*

$$\begin{aligned}(a+b)^n &= \sum_{i=0}^{n} \binom{n}{i} a^{n-i} b^i \\ &= a^n + \ldots + \binom{n}{i} a^{n-i} b^i + \ldots + b^n.\end{aligned}$$

This theorem is only the simplest form of the binomial theorem. There are versions of the theorem when the exponent is a negative integer or a non-integer but in these cases the finite sum is replaced by an infinite series.†

† The more general theorem was discovered by the English mathematician Isaac Newton in the seventeenth century. Details can be found in books on advanced calculus or mathematical analysis – see for example the excellent book R. Courant and F. John, *Introduction to calculus and analysis,* Volume I, Wiley, 1965. No doubt Professor Moriarty's treatise was on this more general theorem.

Examples 12.3.2 Using the binomial theorem we can read off the following identities from Pascal's triangle:

$$\begin{aligned}
(a+b)^2 &= a^2+2ab+b^2;\\
(a+b)^3 &= a^3+3a^2b+3ab^2+b^3;\\
&\vdots\\
(a+b)^7 &= a^7+7a^6b+21a^5b^2+35a^4b^3+35a^3b^4+21a^2b^5+7ab^6+b^7.
\end{aligned}$$

Proof of the binomial theorem Recall (from Definition 5.3.3) that we have adopted the convention that $x^0 = 1$ for all non-zero real numbers x (including $x = 0$).

The proof is by induction on n. Let $P(n)$ be the statement we are required to prove.
Base case: For $n = 0$ this statement reads $(a+b)^0 = \binom{0}{0}a^0b^0$ which is true since $(a+b)^0 = a^0 = b^0 = \binom{0}{0} = 1$.
Inductive step: Suppose now as inductive hypothesis that, for some non-negative integer k, $P(n)$ holds for $n = k$. Then, for all a and b,

$$\begin{aligned}
(a+b)^{k+1} &= (a+b)^k(a+b)\\
&= \sum_{i=0}^{k}\binom{k}{i}a^{k-i}b^i(a+b) \quad \text{by inductive hypothesis}\\
&= \sum_{i=0}^{k}\binom{k}{i}a^{k-i+1}b^i + \sum_{i=0}^{k}\binom{k}{i}a^{k-i}b^{i+1}\\
&= \sum_{i=0}^{k}\binom{k}{i}a^{k-i+1}b^i + \sum_{j=1}^{k+1}\binom{k}{j-1}a^{k+1-j}b^j \quad \text{(writing } j \text{ for } i+1)\\
&= \sum_{i=0}^{k}\binom{k}{i}a^{k+1-i}b^i + \sum_{i=1}^{k+1}\binom{k}{i-1}a^{k+1-i}b^i \quad \text{(writing } i \text{ for } j)\\
&= a^{k+1}b^0 + \sum_{i=1}^{k}\binom{k}{i}a^{k+1-i}b^i + \sum_{i=1}^{k}\binom{k}{i-1}a^{k+1-i}b^i + a^0b^{k+1}\\
&\qquad\qquad \text{since } \tbinom{k}{0} = \tbinom{k}{k} = 1\\
&= a^{k+1} + \sum_{i=1}^{k}\binom{k+1}{i}a^{k+1-i}b^i + b^{k+1} \quad \text{by Proposition 12.2.8}\\
&= \sum_{i=0}^{k+1}\binom{k+1}{i}a^{k+1-i}b^i
\end{aligned}$$

as required to prove $P(k+1)$.
Conclusion: Hence by induction $P(n)$ holds for all positive integers n.

□

This provides an alternative proof of Proposition 12.2.7.

Corollary 12.3.3

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n.$$

Proof Simply put $a = b = 1$ in the binomial theorem. □

Exercises

12.1 Write down the next three rows of Pascal's triangle (page 151) and hence find the coefficient of a^3b^7 in $(a+b)^{10}$.

12.2 Find the coefficient of $a^{98}b^2$ in $(a+b)^{100}$.

12.3 Three people each select a different card from a single pack of 52 distinct cards. How many choices are possible (i) if we record who selected which card, and (ii) if we forget who selected which card?

12.4 Three people each select a main dish from a menu of five items. How many choices are possible (i) if we record who selected which dish (as the waiter should), and (ii) if we ignore who selected which dish (as the chef could)?

12.5 By equating the coefficients of x^n in $(1+x)^{2n} = (1+x)^n(1+x)^n$ prove that

$$\binom{2n}{n} = \sum_{i=0}^{n} \binom{n}{i}^2.$$

12.6 Prove that the product of any n consecutive positive integers is divisible by $n!$.

12.7 Use $(1+i)^n$ to prove that

$$\sum_{0\leqslant 2r\leqslant n}(-1)^r\binom{n}{2r}=\begin{cases}(-4)^k & \text{if } n=4k,\\ (-4)^k & \text{if } n=4k+1,\\ 0 & \text{if } n=4k+2,\\ \frac{1}{2}(-4)^{k+1} & \text{if } n=4k+3.\end{cases}$$

Obtain a similar result for $\displaystyle\sum_{0\leqslant 2r+1\leqslant n}(-1)^r\binom{n}{2r+1}$.

[This question uses complex numbers. The binomial theorem, Theorem 12.3.1, holds also for complex numbers a and b, with the same proof.]

13
Number systems

Numbers originally arose for the purpose of counting. This makes use of the positive integers $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$, often called the natural or counting numbers. In order to write down numbers we need some system of *numerals*, a system of symbols to represent numbers. At first the notation used was simply some sort of tally or repeated mark but in time more elaborate systems of numerals were devised leading eventually to our present Hindu–Arabic decimal system of numerals which makes use of 'cypherization' (i.e. different symbols 1, 2, 3, 4, 5, 6, 7, 8, 9 for different numbers) and a 'place value system' (e.g. in 121 the first '1' denotes 100, the '2' denotes $2 \times 10 = 20$, and the second '1' denotes 1 so that the number represented is the sum $100 + 20 + 1$).

An effective place value system requires a symbol to denote the absence of a positive digit and it is for this purpose that a symbol like '0' first appeared (so that, for example, 102 and 12 represent different numbers). In due course zero was recognized as a number in its own right and then much later negative integers were introduced giving the complete system of integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ in which any two elements may be added or subtracted.

However, numbers are also used for purposes of measurement, originally of lengths but then of many other quantities. Before 1800 B.C. the Babylonians had developed a system of sexagesimal fractions rather like our modern decimals but using a base of 60 rather than 10. These were used until relatively modern times, particularly in astronomy, and indeed are still used in the measurement of angles by degrees, minutes and seconds and time by hours, minutes and seconds. For example if we state that the size of an angle is $18°15'53''$ we are saying that it is $18 + \dfrac{15}{60} + \dfrac{53}{(60)^2} = 18\dfrac{953}{3600}$ degrees. Other ancient cultures had different techniques for dealing with fractions.

The classical Greek mathematicians took the view that the *numbers* were simply the positive integers.† Indeed, they took the view that the essence of all things is explainable in terms of the positive integers and their ratios. The Pythagoreans (founded by Pythagoras about 500 B.C.) assumed at first that any two lengths were *commensurable*, meaning that, given any two line segments, some unit of length could be found so that the length of each of the lines is a whole number times this unit. If then the first line is m units and the second is n units for positive integers m and n, this means that the ratio of the lengths of the two lines is m/n for some positive integers m and n, i.e. it is a rational number.

A simple consequence of Pythagoras's theorem ('the square on the hypotenuse of a right-angled triangle is the sum of the squares on the other two sides') is that the length of the diagonal of a unit square is a number whose square is 2. Some time around 400 B.C. the Pythagoreans discovered that the diagonal and side of a square are *incommensurable* or, equivalently, that there is no rational number whose square is 2. This undermined their view of the place of the positive integers in the world and was kept a secret; there is a story that the person who eventually revealed this secret of the Pythagoreans was executed by drowning. In due course an elegant way round this dilemma was discovered by Eudoxus whose theory of proportion‡ appears in Book Five of the *Elements* of Euclid (about 300 B.C.). This involved a separation of the concepts of 'number' (which meant a positive integer) and 'magnitude' (which meant the length of a line). It was not until the nineteenth century that the two concepts were satisfactorily brought back together again through the work of mathematicians such as Richard Dedekind in Germany.

This book is not the place to go into further details about these later developments. However, it is important that the reader realizes that there are real difficulties in developing a satisfactory system of numbers and numerals.

In this chapter we review what is involved in the algebra of *rational numbers* and go on to prove the classic result that there is no rational square root of 2. A numeral system sufficient to measure all lengths is

† To be precise, the Greeks did not regard 1 as a number. Euclid (Book Seven of the *Elements*) refers to 1 as a *unit* and then defines a *number* to be a multitude composed of units.

‡ See, for example, C.B. Boyer and U.C. Merzbach, *A history of mathematics*, Wiley, Second edition 1989.

provided by *infinite decimals* and we conclude the chapter by discussing this system and indicating some of its subtleties.

Further extensions of the number system are possible. The reader may well be familiar with one possibility: the system of *complex numbers* which first arose in sixteenth century Italy in the solution of cubic and quartic equations (but was not explained properly until the nineteenth century). Complex numbers are not described in this book although a few problems make use of them. The reader will meet them in the course of the study of algebra or advanced calculus.

13.1 The rational numbers

The rational number system provides a set of numbers which can be added and multiplied and which satisfy all the basic algebraic properties of numbers as listed in Properties 2.3.1. The difficulty with the integers is that we cannot always do division. For integers $b \neq 0$ and a we can only find an integer solution of the equation $bx = a$ if b divides a. However, there is always a unique rational number which is a solution. Furthermore the set of rational numbers provide just enough numbers for this purpose: given a rational number q, there are integers $b \neq 0$ and a such that $bq = a$.

A satisfactory definition of rational numbers involves ideas which will be discussed later in this book – such a definition will be indicated in Chapter 22 (see Example 22.3.5). For the moment the reader is asked to accept that a system of rational numbers with the properties described above exists. We now consider how these properties lead to the description of the numbers and their properties in terms of fractions. While doing this it is convenient to make a distinction between a rational number and a fraction: each rational number is representable by a fraction but different fractions may represent the same rational number.

Definition 13.1.1 *We will use the word* $\boxed{\text{fraction}}$ *to denote an expression of the form a/b where a and b are integers and b is non-zero; a is the* $\boxed{\text{numerator}}$ *and b the* $\boxed{\text{denominator}}$ *of the fraction.*

Given integers a and (non-zero) b we say that the fraction a/b represents the $\boxed{\text{rational number}}$ *q which has the property that $bq = a$.*

As a temporary notation we indicate that the rational number q is represented by the fraction a/b by writing

$$q = \langle a/b \rangle.$$

The equation $1x = a$, for a an integer, has an integer solution, namely a, and so the integers can be thought of as a subset of the rational numbers: an integer a can be represented as a fraction by $a = \langle a/1 \rangle$.

Proposition 13.1.2 *Two fractions a_1/b_1 and a_2/b_2 represent the same rational number q, $\langle a_1/b_1 \rangle = \langle a_2/b_2 \rangle = q$, if and only if $a_1 b_2 = a_2 b_1$.*

Proof The fraction a_1/b_1 represents the rational number q_1 such that $b_1 q_1 = a_1$, and the fraction a_2/b_2 represents q_2 such that $b_2 q_2 = a_2$. Using the basic properties of multiplication (commutativity, associativity and cancelling) it follows that

$$\begin{aligned} b_1 q_1 = a_1 \quad &\Leftrightarrow \quad b_1 b_2 q_1 = a_1 b_2, \\ b_2 q_2 = a_2 \quad &\Leftrightarrow \quad b_1 b_2 q_2 = a_2 b_1. \end{aligned}$$

Hence

$$q_1 = q_2 \Leftrightarrow b_1 b_2 q_1 = b_1 b_2 q_2 \Leftrightarrow a_1 b_2 = a_2 b_1.$$

□

Thus for example $\langle 2/3 \rangle = \langle 8/12 \rangle = \langle (-14)/(-21) \rangle$.

By multiplying numerator and denominator by -1 if necessary, we can represent any rational number by a fraction with positive denominator. Then, by dividing the numerator and denominator by their greatest common divisor, we see that every rational number may be written in the form $q = \langle a/b \rangle$, where $b > 0$ and the greatest common divisor $(a, b) = 1$ (see Exercise 11.4).

Definition 13.1.3 *The fraction a/b is* $\boxed{\text{in lowest terms}}$ *when b is positive and a and b are coprime.*

Thus every rational number may be represented by a fraction in lowest terms.

To see how to describe the sum of two rational numbers in terms of fractions suppose that $q_1 = \langle a/b \rangle$ and $q_2 = \langle c/d \rangle$ so that $bq_1 = a$ and $dq_2 = c$ (where a, b, c and d are integers with b and d non-zero). Then $bdq_1 = ad$ and $bdq_2 = bc$ so that (using distributivity) $bd(q_1 + q_2) = ad + bc$ which means that $q_1 + q_2$ is represented by $(ad+bc)/bd$. Similarly, since

$bdq_1q_2 = ac$, q_1q_2 is represented by ac/bd. This discussion motivates the following definition.

Definition 13.1.4 *We define* $\boxed{\text{sum}}$ *and* $\boxed{\text{product}}$ *of two fractions by*

$$\begin{aligned}\frac{a}{b}+\frac{c}{d} &= \frac{ad+bc}{bd},\\ \frac{a}{b}\times\frac{c}{d} &= \frac{ac}{bd}.\end{aligned}$$

Now it is necessary to check that these definitions lead to *well-defined* definitions of addition and multiplication of rational numbers. This procedure is always necessary when we have different expressions for the same object and then make a definition using these expressions. In this case we need to check that if different fractions are used to represent the rational numbers then adding or multiplying the fractions leads to fractions representing the same rational number.

Proposition 13.1.5 *Addition and multiplication of rational numbers are well-defined by the above formulae.*

Constructing a proof. We deal simply with addition. The proof for multiplication is similar and easier. Let us look at what we have to do.

Given	Goal
$\langle a_1/b_1\rangle = \langle a_2/b_2\rangle$	
$\langle c_1/d_1\rangle = \langle c_2/d_2\rangle$	$\langle a_1/b_1 + c_1/d_1\rangle = \langle a_2/b_2 + c_2/d_2\rangle$

When we spell out what these various statements mean, this leads to the following.

Given	Goal
$a_1b_2 = a_2b_1$	$(a_1d_1 + b_1c_1)b_2d_2$
$c_1d_2 = c_2d_1$	$= (a_2d_2 + b_2c_2)b_1d_1$

But now we can obtain the goal from the given statements by some simple algebraic manipulation as follows:

$$\begin{aligned}(a_1d_1+b_1c_1)b_2d_2 &= a_1b_2d_1d_2 + b_1b_2c_1d_2\\ &= a_2b_1d_1d_2 + b_1b_2c_2d_1 \quad \text{by the given statements}\\ &= (a_2d_2+b_2c_2)b_1d_1.\end{aligned}$$

Proof Suppose that $\langle a_1/b_1\rangle = \langle a_2/b_2\rangle$ and $\langle c_1/d_1\rangle = \langle c_2/d_2\rangle$ so that $a_1b_2 = a_2b_1$ and $c_1d_2 = c_2d_1$. Then $\langle a_1/b_1 + c_1/d_1\rangle = \langle(a_1d_1 + b_1c_1)/b_1d_1\rangle$ is the same as $\langle a_2/b_2 + c_2/d_2\rangle = \langle(a_2d_2 + b_2c_2)/b_2d_2\rangle$ since $(a_1d_1 + b_1c_1)b_2d_2 = (a_2d_2 + b_2c_2)b_1d_1$ (using the manipulation shown above). This shows that addition is well-defined.

The proof that multiplication is well-defined is similar. □

Notice that if $q = \langle a/b\rangle$ then $bq = \langle b/1 \times a/b\rangle = \langle ab/b\rangle = \langle a/1\rangle = a$ and so this definition of multiplication is consistent with Definition 13.1.1.

From now on we abandon the notation $\langle a/b\rangle$ and simply write a/b for the rational number represented by the fraction a/b as usual. Thus $a_1/b_1 = a_2/b_2$ if and only if $a_1b_2 = a_2b_1$. The set of rational numbers is denoted by the symbol $\mathbb{Q}$.

The above discussion has described how addition and multiplication are defined in $\mathbb{Q}$. But of course subtraction of any element and division by non-zero elements are always possible in $\mathbb{Q}$ since

$$\frac{a}{b} - \frac{c}{d} = \frac{a}{b} + \frac{-c}{d}$$

and, for $c \neq 0$,

$$\frac{a}{b}\Big/\frac{c}{d} = \frac{a}{b} \times \frac{d}{c}.$$

13.2 The irrationality of $\sqrt{2}$

The previous section has described the rational numbers and their familiar arithmetic. It was suggested above that a good number system should enable us to represent the length of any line. Pythagoras's theorem tells us that the length of the diagonal of a unit square is a number whose square is 2, usually denoted by $\sqrt{2}$. The next result tells us that there is no such rational number.

Theorem 13.2.1 *There does not exist a rational number whose square is* 2.

Constructing a proof. Since every non-zero rational number can be written in the form a/b where a and b are non-zero integers we can write the result we are trying to prove as follows:

$$\nexists a, b \in \mathbb{Z} - \{0\},\ 2 = (a/b)^2.$$

Whenever a result takes the form of stating that something does not happen it is very difficult to envisage a direct proof and this suggests very strongly that we need a proof by contradiction. Remember that this requires us to suppose that our goal is false and then to seek a contradiction. The negation of the goal is

$$\exists a, b \in \mathbb{Z} - \{0\},\ 2 = (a/b)^2.$$

We use this existential statement by selecting elements a and b so that the predicate is true. This gives the following strategy.

Given	Goal
$a, b \in \mathbb{Z} - \{0\}$ $2 = (a/b)^2$	contradiction

We can rewrite this as follows.

Given	Goal
$a, b \in \mathbb{Z} - \{0\}$ $a^2 = 2b^2$	contradiction

Thus we see that the theorem we are trying to prove is really one about the integer solutions to an equation:

The only integer solution to the equation $m^2 = 2n^2$ is the trivial one $m = 0$, $n = 0$.

The one method we have used for proving that an equation has no integer solutions (see for example Proposition 4.1.1) is to use the idea of divisibility. So let us try that here. Given that $a^2 = 2b^2$ then we know that a^2 must be an even number. What does this tell us about a?

We know that a is either even or odd and it follows by a proof by contradiction that in fact a must be even (see Problems I, Question 7).

Returning to our argument, we have seen that $a^2 = 2b^2$ implies that a is even. Hence we can write $a = 2a_1$. Substituting this into $a^2 = 2b^2$ we get $4a_1^2 = 2b^2$ so that $2a_1^2 = b^2$. Hence b^2 is even and so by the same argument b is even. So we have concluded that, if $a^2 = 2b^2$, then the numbers a and b must both be even.

So what? This in itself is no contradiction but it does suggest how we might get one, for if a and b are both even this means that the fraction a/b which we began with is not written in its lowest terms since a and b have the common factor 2. But we know that every rational number can be represented by a fraction in lowest terms. So had we thought to start by insisting that a/b be the lowest terms fraction representing a square root of 2 then we would now have obtained a contradiction. So we go back and place the requirement that $(a, b) = 1$ on a and b.

This gives the following effective strategy for proving the theorem by contradiction.

Given	Goal
$a, b \in \mathbb{Z} - \{0\}$	
$(a,b) = 1$	contradiction
$a^2 = 2b^2$	[i.e. $(a,b) = 1$ and $(a,b) \neq 1$]

Proof Suppose for contradiction that there is a rational number q such that $q^2 = 2$. Write q as a fraction in lowest terms, $q = a/b$, so that a and b are integers such that $(a,b) = 1$. Now $q^2 = 2 \Rightarrow a^2/b^2 = 2 \Rightarrow a^2 = 2b^2 \Rightarrow a^2$ is even $\Rightarrow a$ is even $\Rightarrow a = 2a_1$ for some $a_1 \in \mathbb{Z}$. But then $a^2 = 2b^2 \Rightarrow 4a_1^2 = 2b^2 \Rightarrow 2a_1^2 = b^2 \Rightarrow b^2$ is even $\Rightarrow b$ is even. Hence 2 divides a and 2 divides b implying that $(a,b) \neq 1$. But this contradicts the choice of a and b and so provides the required contradiction.

Hence there does not exist a rational number whose square is 2. □

13.3 Real numbers and infinite decimals

The result in the previous section demonstrates that the rational number system is not adequate to represent all lengths in terms of some standard unit length since the square of the length of the diagonal of a unit square is 2 and there is no rational number whose square is 2. The *real numbers* provide a further extension of the number system which is adequate to represent the lengths of all line segments. A real number which is not a rational number is said to be *irrational*. So Theorem 13.2.1 can be reformulated as the statement '$\sqrt{2}$ is irrational'.

The definition of a real number is beyond the scope of this book since it requires a full treatment of the theory of limits of sequences referred to briefly in Chapter 8. This material can be found in any book on infinite sequences and series or mathematical analysis.†

The notation used to describe real numbers is that of *infinite decimals*. Thus every non-negative real number may be represented by an infinite decimal

$$a_0.a_1a_2\ldots a_i\ldots, \quad \text{where } a_i \in \mathbb{Z}^{\geqslant} \text{ and } 0 \leqslant a_i \leqslant 9 \text{ for } i \geqslant 1$$

and every infinite decimal represents a real number. The aim in this sec-

† See for example R. Haggerty, *Fundamentals of mathematical analysis*, Addison-Wesley, Second edition 1993.

tion is describe how the real number represented by an infinite decimal may be specified.

First of all, notice that a finite decimal $a_0.a_1a_2\dots a_n$ (i.e. an infinite decimal with $a_i = 0$ for $i > n$) is equal to a rational number:

$$\begin{aligned} & a_0.a_1a_2\dots a_n \\ = \quad & a_0 + a_1/10 + a_2/100 + \dots + a_n/(10)^n \\ = \quad & (a_0(10)^n + a_1(10)^{n-1} + a_2(10)^{n-2} + \dots + a_n)/(10)^n. \end{aligned}$$

So the set of finite decimals is a subset of the set of rational numbers.

The intuitive idea of an infinite decimal is that as we take more and more decimal places we get finite decimals which provide a better and better approximation to the number. Before we give a precise definition let us consider an example.

Example 13.3.1 We have already proved that there is no rational number whose square is 2. Let us consider how to find an infinite decimal representing the positive real number whose square is 2. We construct the integers $a_0, a_1, \dots$ making up the decimal by an inductive process. The key observation which enables us to do this is that, for non-negative real numbers a and b, $a < b$ if and only if $a^2 < b^2$ (see Exercise 4.6).

Step 0: To find the integer part we find the largest integer whose square is less than 2. Since $1^2 = 1 < 2 < 2^2 = 4$ we know that $1 < \sqrt{2} < 2$ and so we put $a_0 = 1$.

Step 1: To find the first entry after the decimal point we square each of the numbers 1.0, $1.1 = 11/10$, $1.2 = 12/10$, ..., $1.9 = 19/10$ in turn and find the largest of them whose square is less than 2. (Notice that we cannot get equality since 2 is not a rational square). Since

$$(14/10)^2 = 196/100 < 2 < (15/10)^2 = 225/100$$

we know that $1.4 < \sqrt{2} < 1.5$ and so put $a_1 = 4$.

Step 2: For the second decimal place, since

$$(141/100)^2 = 19881/10000 < 2 < (142/100)^2 = 20164/10000$$

so that $1.41 < \sqrt{2} < 1.42$ we put $a_2 = 1$.

This process continues indefinitely giving an infinite decimal whose first few decimal places are 1.41421356237309504880 – which means that

$$1.41421356237309504880 < \sqrt{2} < 1.41421356237309504881.$$

However, there is no apparent pattern to the numbers which emerge. By Step n we have found a finite decimal $a_0.a_1a_2\dots a_n$ such that

$$(a_0.a_1a_2\dots a_n)^2 < 2 < (a_0.a_1a_2\dots a_n + 1/(10)^n)^2$$

which means that

$$a_0.a_1a_2\dots a_n < \sqrt{2} < a_0.a_1a_2\dots a_n + 1/(10)^n.$$

For example, saying that the first seven decimal places of $\sqrt{2}$ are given by 1.4142135 means that

$$1.4142135 < \sqrt{2} < 1.4142136$$

and, if we do the arithmetic, we find that it is indeed the case that

$$1.4142135^2 = 1.99999982358225 < 2 < 1.4142136^2 = 2.00000010642496.$$

This example illustrates the formal definition.

Definition 13.3.2 *Given an infinite decimal $a_0.a_1a_2\dots a_i\dots$, we say that it represents a real number a, written $a = a_0.a_1a_2\dots a_i\dots$, when*

$$\boxed{a_0.a_1a_2\dots a_n \leqslant a \leqslant a_0.a_1a_2\dots a_n + 1/(10)^n \text{ for each } n \in \mathbb{Z}^+.}$$

Notice that we use '$\leqslant$' rather than '$<$' in this definition. This is necessary so that all rational numbers may be represented by an infinite decimal (possibly with $a_i = 0$ for $i > n$, some n).

The statement that every infinite decimal represents *some* real number is one version of the *completeness* or *continuity axiom* for the real numbers.

To see that an infinite decimal is equal to a *unique* real number we observe that the inequalities in the definition uniquely specify the number a. Given two distinct real numbers a and a' we have $a - a' \neq 0$. Now we choose an integer n so that $1/(10)^n < |a - a'|$ by taking $n > 1/|a - a'|$ (for then $1/(10)^n < 1/n < |a - a'|$). [The fact that for each real number there is a greater integer is sometimes called the *Archimedean axiom* for the real numbers.] This means that infinite decimals representing a and a' will certainly differ by the nth decimal place.

Example 13.3.3 Let us consider which real number is represented by the infinite decimal $0.999\dots99\dots$ each of whose entries after the decimal point is 9. An infinite decimal like this is called a *recurring* or repeating decimal, and we use the notation $0.\dot{9}$ to represent it with a dot over the integer which recurs. Recurring decimals may have a block of digits recurring, for example 7.320008140081400814.... In this case we indicate the repeating block by placing a dot over the first and last digits in the block, viz. $7.320\dot{0}081\dot{4}$.†

The finite decimals $0.99\ldots9$ are certainly less than 1. Indeed,

$$0.\underbrace{99\ldots9}_{n} = \underbrace{99\ldots9}_{n}/(10)^n = \frac{(10)^n - 1}{(10)^n} = 1 - \frac{1}{(10)^n}.$$

Intuition may suggest that this means that the infinite decimal $0.\dot{9}$ is also less than 1 and this is a common misconception. I have known people become quite belligerent in their defence of this view. This is not an effective way of defending any point of view but emotion is particularly inappropriate in a mathematical argument! As always, if in doubt apply Humpty Dumpty's principle and examine what the words have been declared to mean. So we return to the definition. Let $a = 0.\dot{9}$. From the definition,

$$\begin{aligned} a = 0.\dot{9} \quad &\Leftrightarrow \quad 1 - 1/(10)^n \leqslant a \leqslant 1 \\ &\Leftrightarrow \quad 0 \leqslant 1 - a \leqslant 1/(10)^n \end{aligned} \tag{13.1}$$

for all positive integers n.

Thus $1-a > 0$ or $1-a = 0$. We eliminate the first of these possibilities by contradiction. If x is a positive number then as above we can find a positive integer k such that $1/(10)^k < x$ by taking $k > 1/x$. Applying this to $x = 1-a$ demonstrates that if $1-a > 0$ then $1-a$ does not satisfy the condition (13.1) for $n = k$, giving a contradiction. The conclusion must be that $1 - a$ is not positive and so $1 - a = 0$ or in other words $a = 1$. This proves that

$$\boxed{0.\dot{9} = 1.}$$

If this result still worries you (and after all why not, since it is based on Definition 13.3.2 which is rather sophisticated?) then you may feel happier when you remember the most familiar recurring decimal, $1/3 = 0.\dot{3}$, which you probably are comfortable with. Multiplying by 3 gives $1 = 0.\dot{9}$.

† Alternatively, we place a dot over each digit which recurs, $7.320\dot{0}\dot{0}\dot{8}\dot{1}\dot{4}$ or a line over the repeating block, $7.320\overline{00814}$.

Recurring decimals and rationals

The arithmetic of infinite decimals is quite complicated.† However, multiplication by 10 is easy for

$$\begin{aligned} 10 \times a_0.a_1a_2\ldots a_n &= 10(a_0 + a_1/10 + a_2/100 + \ldots + a_n/(10)^n) \\ &= 10a_0 + a_1 + a_2/10 + \ldots + a_n/(10)^{n-1} \\ &= (10a_0 + a_1).a_2\ldots a_n. \end{aligned}$$

So multiplication of a finite decimal by 10 corresponds to moving the decimal point one place to the right. This means that the same is true of infinite decimals since

$$\begin{aligned} &a_0.a_1a_2\ldots a_n \leqslant a \leqslant a_0.a_1a_2\ldots a_n + 1/(10)^n \\ &\quad\Leftrightarrow\quad (10a_0 + a_1).a_2\ldots a_n \leqslant 10a \leqslant (10a_0 + a_1).a_2\ldots a_n + 1/(10)^{n-1}. \end{aligned}$$

This observation enables us to prove the following result.

Theorem 13.3.4 *A recurring infinite decimal is equal to a rational number.*

The idea of the proof of this result is very simple but writing out a general proof leads to great notational complexity. The proof consists of an algorithm (or procedure) for finding the rational number. If the length of the recurring block is k we multiply through by 10^k and subtract the number we started with. The result is a finite decimal and so a rational number. This enables us to write the recurring decimal as a rational number. Rather than writing out a general proof the method is illustrated by giving an example.

Example 13.3.5 To find the rational number $12.79317317317317\ldots = 12.79\dot{3}1\dot{7}$.

Solution Let $x = 12.79\dot{3}1\dot{7}$. Since the length of the recurring part is 3 we multiply through by 10^3 giving $1000x = 12793.\dot{1}7\dot{3}$ (since the decimal point moves three places to the right) $= 12793.17\dot{3}1\dot{7}$. From the definition of the value of an infinite decimal we can split off an initial finite decimal:

$$x = 12.79 + 0.00\dot{3}1\dot{7},$$

† An excellent discussion of the arithmetic of infinite decimals, a topic bypassed in most treatments of infinite decimals, may be found in A. Gardiner, *Infinite processes, background to analysis,* Springer-Verlag, 1982.

$$1000x = 12793.17 + 0.00\dot{3}1\dot{7}.$$

Subtracting we obtain

$$999x = 12793.17 - 12.79 = 12780.38 = 1278038/100$$

and so $x = 1278038/99900$, a rational number as required. □

We will see later on that the converse of this theorem is also true (see Problems IV, Question 5). This means that the rational numbers correspond precisely to the recurring decimals (a finite decimal can be thought of as a recurring infinite decimal: $a_0.a_1 \ldots a_n = a_0.a_1 \ldots a_n\dot{0}$).

Exercises

13.1 Prove that there does not exist a rational number whose square is 3.
[You may assume that a^2 is divisible by 3 if and only if a is divisible by 3 (proved later as Proposition 15.2.1).]

13.2 Find the first two decimal places of an infinite decimal representing $\sqrt{3}$.

13.3 Try to mimic the proof of Theorem 13.2.1 to show that there does not exist a rational number whose square is 4. Where does this go wrong?

13.4 Prove from Definition 13.3.2 that $1/3 = 0.\dot{3}$.

13.5 Find the rational number equal to the recurring infinite decimal $1.778\dot{2}34\dot{5}$.

14
Counting infinite sets

In Chapter 10 a non-empty set X was defined to have cardinality n, where n is a positive integer, when there is a bijection $\mathbb{N}_n \to X$ from the standard set $\mathbb{N}_n = \{1, 2, \ldots, n\}$. Such sets, together with the empty set, were defined as *finite* and all others as *infinite*.

We saw (Exercise 10.1) that, given a finite set X, another set Y is also finite of the same cardinality if and only if there is a bijection $X \to Y$. In this case we say that X and Y are *equipotent*. A bijection $X \to Y$ guarantees that X and Y have the same cardinality.

These ideas may be extended to infinite sets. We can think of any two sets as having the same cardinality if they are equipotent so that there is a bijection between them. We can then describe this cardinality by introducing additional standard sets. For example, when we embark on counting a set X we assign a distinct element of the set to each positive integer in turn. If we exhaust the elements on reaching the integer n then we have constructed a bijection $\mathbb{N}_n \to X$ and so the set is finite with cardinality n. If we never exhaust the elements we may nevertheless reach each element of the set eventually so that we have a bijection $\mathbb{Z}^+ \to X$. In this case the set is equipotent to the standard set $\mathbb{Z}^+$ and we say that the set, although infinite, is *denumerable*.

In this chapter we will explore the properties of infinite sets, considering which properties of finite sets can be extended to such sets. It will become clear that not all do so and this should help explain why such care was needed in developing the theory of finite sets. The fact that when we come to infinite sets various 'obvious' results are no longer true suggests that these results may not be quite so obvious after all! In particular we will see that care must be taken when comparing the 'size' of infinite sets but that it is possible to do this. It is even possible in certain cases to use 'counting arguments' with infinite sets.

The material in this chapter is somewhat harder than most of the rest of the book. It is not used subsequently in the book but is included to provide a context for the earlier material on finite sets and because the ideas are so striking and should be part of any general mathematical education.

14.1 Countable sets

Definition 14.1.1 *Two sets X and Y are* $\boxed{\text{equipotent}}$ *if there is a bijection $X \to Y$.*

Thus X is finite of cardinality $n \in \mathbb{Z}^+$ if and only if X is equipotent to $\mathbb{N}_n$.

Definition 14.1.2 *A set X is said to be* $\boxed{\text{denumerable}}$ *or* enumerable *if there is a bijection $\mathbb{Z}^+ \to X$, so that it is equipotent to $\mathbb{Z}^+$. A set is* $\boxed{\text{countable}}$ *if either it is finite or it is denumerable. A set is* $\boxed{\text{uncountable}}$ *if it is not countable.*

If the set X is denumerable then it is said to have $\boxed{\text{cardinality}}$ $\aleph_0$ *and this is written* $\boxed{|X| = \aleph_0}$.

The notation for the cardinality of a denumerable set uses the Hebrew letter aleph and '$\aleph_0$' is usually read 'aleph null'.

Conventions vary a little and in some books the word 'countable' is used where we are using 'denumerable' or conversely.

If a set X is denumerable then we can list the elements of the set, but the list is infinite. Given a bijection $f: \mathbb{Z}^+ \to X$,

$$X = \{x_1, x_2, \ldots, x_n, \ldots\}$$

where we write $x_n = f(n)$.

Examples 14.1.3 (a) $\mathbb{Z}^+$ is denumerable by taking the identity function $\mathbb{Z}^+ \to \mathbb{Z}^+$. This gives the listing $\mathbb{Z}^+ = \{1, 2, \ldots, n, \ldots\}$.

(b) The set of non-negative integers $\mathbb{Z}^{\geqslant}$ is denumerable since a bijection $\mathbb{Z}^+ \to \mathbb{Z}^{\geqslant}$ is defined by $n \mapsto n-1$. This gives the listing $\mathbb{Z}^{\geqslant} = \{0, 1, \ldots, n-1, \ldots\}$.

(c) The set of all integers $\mathbb{Z}$ is denumerable, for a bijection $\mathbb{Z}^+ \to \mathbb{Z}$ may be defined by

$$n \mapsto \begin{cases} -(n-1)/2 & \text{if } n \text{ is odd,} \\ n/2 & \text{if } n \text{ is even.} \end{cases}$$

This gives the listing $\mathbb{Z} = \{0, 1, -1, 2, -2, \ldots, n, -n, \ldots\}$.

(d) The set of positive even integers $X = \{\, 2n \mid n \in \mathbb{Z}^+ \,\}$ is denumerable, for a bijection $\mathbb{Z}^+ \to X$ may be given by $n \mapsto 2n$. This gives the listing $\{2, 4, \ldots, 2n, \ldots\}$.

(e) The set of positive perfect squares $X = \{\, n^2 \mid n \in \mathbb{Z}^+ \,\}$ is denumerable, for a bijection $\mathbb{Z}^+ \to X$ may be given by $n \mapsto n^2$. This gives the listing $X = \{1, 4, \ldots, n^2, \ldots\}$.

'Paradoxes' of the infinite

Recall from Exercise 11.5 that a finite set can never have the same cardinality as a proper subset: the cardinality of a proper subset of a finite set is necessarily smaller. However, the above examples demonstrate that a denumerable set can have equipotent proper subsets. Example 14.1.3(d) shows that the set of positive integers and the positive even integers are equipotent and so, in that sense, there are the 'same number' of even integers as integers even though it appears that only half the integers are even!

The first reference to this 'paradox' appears in the work of Galileo Galilei.† He observes that, despite the fact that the further you go along the sequence of positive integers the scarcer the perfect squares become, we must say that there are as many squares as there are positive integers because of the correspondence between the positive integers and the squares given by the bijection $n \mapsto n^2$ (Example 14.1.3(e)). He suggests that the idea of one infinite quantity being greater than another has no meaning.

These ideas seem not to have been developed further until 1850 when a book by the unconventional Czech priest Bernhard Bolzano was published posthumously.‡ He showed that there were many examples of bijections between infinite sets and proper subsets. He also suggested that

† In his 1638 book *Discorsi e dimostrazioni matematiche intorno à due nuove scienze* (written, remarkably for the period, in Italian rather than Latin and often referred to by its English title *The two new sciences*).

‡ *Paradoxien des Unendlichen* published in English as *Paradoxes of the infinite* (edited by D.A. Steele) in 1950.

nevertheless certain infinite sets are in a real sense bigger than others, anticipating the work of Cantor which we will come to shortly.

In 1872 the German mathematician Richard Dedekind defined an infinite set to be one which is equipotent to a proper subset of itself.† We now observe that this is equivalent to our definition of infinite set.

Theorem 14.1.4 (Dedekind) *A set X is infinite if and only if it is equipotent to a proper subset of itself.*

Proof If X is equipotent to a proper set then, by Exercise 11.5, X is not finite and so must be infinite.

For the converse, first of all we prove that an infinite set X must contain a denumerable subset. We define this set $A = \{a_1, a_2, \ldots, a_n \ldots\}$ inductively.
Base case: Choose any element of the set X as a_1.
Inductive step: Suppose that, for some $k \geqslant 1$, a subset of k distinct elements $A_k = \{\, a_n \mid 1 \leqslant n \leqslant k \,\}$ has been defined. Choose any element of $X - A_k$ [non-empty since A_k is finite and X is infinite] as a_{k+1}.
This defines a denumerable subset $A = \{a_1, a_2, \ldots, a_n, \ldots\} \subseteq X$.

We may now define a bijection $f\colon X \to X - \{a_1\}$ onto a proper subset of X, as required, by

$$f(x) = \begin{cases} a_{n+1} & \text{if } x = a_n \in A, \\ x & \text{if } x \notin A. \end{cases}$$

Hence X and the proper subset $X - \{a_1\}$ are equipotent. □

14.2 Denumerable sets

For two finite sets, their cardinality determines whether or not they are equipotent, i.e. whether or not there is a bijection between them (Exercise 10.1). This property extends to sets of cardinality $\aleph_0$, denumerable sets, as follows.

Proposition 14.2.1 *Given a denumerable set X, a set Y is also denumerable if and only if it is equipotent to X.*

Proof Since X is denumerable there is a bijection $f\colon \mathbb{Z}^+ \to X$.

† In his book *Stetigkeit und irrationale Zahlen.*

Suppose that Y is also denumerable. Then there is a bijection $g: \mathbb{Z}^+ \to Y$. Since a bijection is invertible we can now deduce that X and Y are equipotent by writing down the bijection $g \circ f^{-1}: X \to \mathbb{Z}^+ \to Y$.

Conversely, if X and Y are equipotent then there is a bijection $h: X \to Y$. We can deduce that Y is denumerable by writing down the bijection $h \circ f: \mathbb{Z}^+ \to X \to Y$. □

When we form unions or products of finite sets we normally obtain bigger sets. One aspect of the unexpected behaviour of infinite sets is that for them this is not the case.

Proposition 14.2.2 *If A and B are denumerable sets then so is their union $A \cup B$.*

This is left as an exercise.

Proposition 14.2.3 *If A and B are denumerable sets then so is their Cartesian product, $A \times B$.*

Constructing a proof. Since A and B are denumerable we can list their elements: $A = \{a_1, a_2, \ldots, a_m, \ldots\}$ and $B = \{b_1, b_2, \ldots, b_n, \ldots\}$. Using these listings we can display the elements of $A \times B$ in a double array.

$$\begin{array}{lllll}
\vdots & & & & \\
(a_1, b_4) & \vdots & & & \\
(a_1, b_3) & (a_2, b_3) & \cdots & & \\
(a_1, b_2) & (a_2, b_2) & (a_3, b_2) & \cdots & \\
(a_1, b_1) & (a_2, b_1) & (a_3, b_1) & (a_4, b_1) & \cdots
\end{array}$$

If we start to count these elements by going along the rows, which would be the normal way of counting a finite array (consider for example the solution to Exercise 10.3) then we will never get to the end of the first row and so no element in the second row will be reached. One way of ensuring† that we do eventually reach every element is to count along

† This should be contrasted with the situation for finite sets. For a finite set it doesn't matter how we start: the addition principle ensures that we will complete the count and Proposition 10.1.3 ensures that the answer will always be the same.

the diagonals in the following pattern.

$$\begin{array}{ccccc}
\vdots & & & & \\
10 & \vdots & & & \\
6 & 9 & \cdots & & \\
3 & 5 & 8 & \cdots & \\
1 & 2 & 4 & 7 & \cdots
\end{array}$$

If we count in this way the element (a_m, b_n) will occur as the nth element in diagonal number $m+n-1$. So (since there are k elements in the kth diagonal) it will be element number

$$1 + 2 + \ldots + (m + n - 2) + n = \tfrac{1}{2}(m + n - 2)(m + n - 1) + n$$

(using Proposition 5.3.1).

Proof Let $A = \{a_1, a_2, \ldots, a_m \ldots\}$ and $B = \{b_1, b_2, \ldots, b_n, \ldots\}$. Then a bijection $A \times B \to \mathbb{Z}^+$ may be defined by

$$(a_m, b_n) \mapsto \tfrac{1}{2}(m + n - 2)(m + n - 1) + n.$$

□

Corollary 14.2.4 *If A is a denumerable set then so is A^n for every positive integer n.*

Proof This is an easy induction proof. The proposition gives the inductive step. □

Proposition 14.2.5 *A subset of a denumerable set is countable (i.e. either finite or denumerable).*

Constructing a proof. As with finite sets we can prove statements about general denumerable sets by first proving them for the standard denumerable set $\mathbb{Z}^+$.

The conclusion of this result is an 'or' statement and so we use the standard strategy introduced in Chapter 4: we take a non-finite subset, A, of $\mathbb{Z}^+$ in other words an infinite subset, and verify that it must be denumerable. We can construct a bijection $\mathbb{Z}^+ \to A$ inductively essentially by listing the elements of A in increasing order starting with the least.

Proof (a) We begin by considering subsets of $\mathbb{Z}^+$, the set of positive integers. Suppose that A is a infinite subset of $\mathbb{Z}^+$. Then to prove that A is denumerable we must construct a bijection $f\colon \mathbb{Z}^+ \to A$. We define $f(n)$ inductively as follows.

Base case: Let $f(1)$ be the least element of the set A. [Recall that by the well-ordering principle (see Example 11.2.2(c)) any non-empty set of positive integers has a least element.]

Inductive step: Suppose that $f(k)$ has been defined for some $k \geqslant 1$. Let $f(k+1)$ be the least element of the set $\{\, a \in A \mid a > f(k) \,\}$. [This set is non-empty since A is infinite.]

This defines a function $f\colon \mathbb{Z}^+ \to A$.

This function is an injection since $f(k) < f(k+1)$ for all $k \geqslant 1$.

To see that f is a surjection notice that $f(n) \geqslant n$ (easily proved by induction). Given an element $a \in A$ then let m be the least integer such that $f(m) \geqslant a$ [we must have $m \leqslant a$ since $f(a) \geqslant a$]. Then by the definition of f we have $f(m) = a$.

Thus we have constructed a bijection $f\colon \mathbb{Z}^+ \to A$ showing that A is denumerable.

Hence each subset of $\mathbb{Z}^+$ is finite or denumerable.

(b) Now consider the general case. Let A be a subset of a denumerable set X. Since X is denumerable there is a bijection $g\colon \mathbb{Z}^+ \to X$. Then $\overleftarrow{g}(A)$ is a subset of $\mathbb{Z}^+$ and so finite or denumerable. But the restriction of g gives a bijection $\overleftarrow{g}(A) \to A$ and so A is finite or denumerable. □

Theorem 14.2.6 (Cantor (1874)) *The set of rationals* $\mathbb{Q}$ *is denumerable.*

Proof We may define a function

$$f\colon \mathbb{Q} \to \mathbb{Z} \times \mathbb{Z}^+$$

by $f(q) = (m,n)$ where m/n is the fraction representing q in its lowest terms (i.e. $q = m/n$, $n > 0$ and $\gcd(m,n) = 1$). This is clearly an injection. Its image is an infinite subset of the denumerable set $\mathbb{Z} \times \mathbb{Z}^+$ and so denumerable. Hence, since f defines a bijection onto its image, $\mathbb{Q}$ is denumerable. □

14.3 Uncountable sets

In the previous section we have seen that surprisingly many infinite sets are denumerable and indeed the examples there support Galileo's suggestion that we can never say that one infinite set is larger than another. However, this is not the case. There are uncountable sets and we can give meaning to the statement that these are larger than the countable sets.

In 1874, Georg Cantor published a remarkable paper in which he proved not only that the set of rational numbers is denumerable but also that there is a hierarchy of infinite sets ordered by size (or cardinality).

Theorem 14.3.1 (Cantor (1874)) *The set of real numbers $\mathbb{R}$ is uncountable.*

Constructing a proof. Notice that the statement that $\mathbb{R}$ is uncountable is a non-existence statement: there does not exist a bijection $\mathbb{Z}^+ \to \mathbb{R}$. The method is to show how given any function $\mathbb{Z}^+ \to \mathbb{R}$ we can construct an element of $\mathbb{R}$ which is not in the image of the function, not a value, so that the function is not a surjection and so not a bijection.

Each element $x \in \mathbb{R}$ may be represented as an infinite decimal

$$x = a_0.a_1a_2 \dots a_n \dots$$

where $a_i \in \mathbb{Z}$ and, for $i \geqslant 1$, $0 \leqslant a_i \leqslant 9$. Some numbers (such as $(1.\dot{0} = 0.\dot{9})$ have two decimal expansions, one ending in recurring 0's (and so a finite decimal) and one ending in recurring 9's (see Problems III, Questions 26 and 27). For these we always use the first representation.

The method is now to start with a function $\mathbb{Z}^+ \to \mathbb{R}$ and to list the values of the function as infinite decimals. Having done this we have a beautifully simple, but ingenious, method (known as 'Cantor's diagonal process') for writing down an infinite decimal not in the list and not representing the same real number as any member of the list. This infinite decimal represents a real number which is not a value of the function.

Proof Suppose that $f: \mathbb{Z}^+ \to \mathbb{R}$ is a function. Let

$$f(m) = a_{m0}.a_{m1}a_{m2} \dots a_{mn} \dots$$

as an infinite decimal (not ending in recurring 9's).

Put $b = 0.b_1b_2 \dots b_n \dots$ where

$$b_n = \begin{cases} 1 & \text{if } a_{nn} = 0, \\ 0 & \text{if } a_{nn} \neq 0. \end{cases}$$

Notice that this infinite decimal representing b certainly does not end in recurring 9's. Furthermore, for each n, the nth decimal place of b, namely b_n, certainly differs from the nth decimal place of $f(n)$, namely a_{nn}. So $b \neq f(n)$. Since we have constructed an element of $\mathbb{R}$ which is not a value of f this shows that f is not a surjection. Hence $\mathbb{R}$ is not denumerable. □

This result shows that the set of real numbers is in a real sense larger than the set of integers (or the set of rationals). We can make this precise as follows.

Definition 14.3.2 *Two sets X and Y have the same* cardinality, *written* $\boxed{|X| = |Y|}$, *if they are equipotent, i.e. there is a bijection $X \to Y$. If there is an injection $X \to Y$ then we write* $\boxed{|X| \leqslant |Y|}$. *We write* $\boxed{|X| < |Y|}$ *to mean that $|X| \leqslant |Y|$ and $|X| \neq |Y|$ and say that X has smaller cardinality than Y.*

Thus Theorem 14.3.1 may be stated as $|\mathbb{R}| > |\mathbb{Z}^+|$ or $|\mathbb{R}| > \aleph_0$. One natural question now is whether there is a set intermediate in size between $\mathbb{Z}^+$ and $\mathbb{R}$. In 1878, Cantor conjectured that such sets did not exist, a statement which became known as the continuum hypothesis. Here the word 'continuum' is simply a name for the set of real numbers $\mathbb{R}$.

The continuum hypothesis. *Every uncountable set of real numbers has the same cardinality as the set of all real numbers:*

$$X \subseteq \mathbb{R} \text{ and } |X| > \aleph_0 \Rightarrow |X| = |\mathbb{R}|.$$

In 1963 the United States mathematician Paul Cohen proved the remarkable result that it is not possible either to prove or to disprove this hypothesis starting from the generally accepted assumptions (axioms) of set theory. This means that there are two alternative versions of set theory, one in which the continuum hypothesis is true and one in which it is false.

The other natural question arising from Theorem 14.3.1 is whether there is a set larger than $\mathbb{R}$. Cantor was able to show that there is such a set and indeed that, given any set, its power set is necessarily larger.

Theorem 14.3.3 *For any set X, $|X| < |\mathcal{P}(X)|$.*

Constructing a proof. Recall that $\mathcal{P}(X)$, the power set of X, is the set of all subsets of X (Definition 6.3.1).

It is easy to define an injection $X \to \mathcal{P}(X)$ by sending each element to the corresponding singleton subset. The interesting part is in proving that there is no bijection $X \to \mathcal{P}(X)$. The method is similar to that of Theorem 14.3.1: starting from any function $f: X \to \mathcal{P}(X)$ we construct an element of $\mathcal{P}(X)$, in other words a subset of X, which is not a value of f. It is easy to describe such an element but the reason why it is not a value is quite subtle.

Proof An injection $X \to \mathcal{P}(X)$ may be defined by $x \mapsto \{x\}$ and so $|X| \leqslant |\mathcal{P}(X)|$.

We prove that there is no bijection $X \to \mathcal{P}(X)$ by showing that any function $X \to \mathcal{P}(X)$ is not a surjection. Suppose that $f: X \to \mathcal{P}(X)$ is a function. We prove that this function is not a surjection and so not a bijection by exhibiting an element of $\mathcal{P}(X)$ which is not a value. Define

$$A = \{ x \in X \mid x \notin f(x) \}.$$

[Notice that, for each $x \in X$, $f(x) \in \mathcal{P}(X)$ is a subset of X and so either $x \in f(x)$ or $x \notin f(x)$. Thus '$x \notin f(x)$' is a predicate with a free variable taking values in X. The set of elements of X for which this predicate is true is a well-defined subset of X: this is the set A.]

To prove that A is not in the image of the function f we suppose for contradiction that it is, say $A = f(a)$ for some $a \in X$. Then A is a subset of X and a is an element of X, and so either $a \in A$ or $a \notin A$. We consider the possibilities in turn.

If $a \in A$ then, referring back to the predicate defining A, $a \notin f(a)$, which can be written $a \notin A$ contradicting our assumption that $a \in A$.

On the other hand, if $a \notin A$, then it does not satisfy the predicate defining A and so $a \in f(a)$, which can be written $a \in A$ contradicting our assumption that $a \notin A$.

So in either case we have a contradiction and so our initial assumption that A is in the image of f must be false and so f is not a surjection.

Hence, since there is no bijection $X \to \mathcal{P}(X)$ but there is an injection, $|X| < |\mathcal{P}(X)|$. □

This is probably the most subtle proof in this book. However, once you have understood it, it appears to have the same glorious inevitability as the great Euclidean proofs by contradiction.

The most important result on finite sets was the pigeonhole principle (Theorem 11.1.2). This result does still hold for infinite sets.

Theorem 14.3.4 (Cantor–Schröder–Bernstein theorem) *Suppose that X and Y are non-empty sets such that $|X| > |Y|$. Then any function $f: X \to Y$ is not an injection, i.e. there exist distinct elements x_1 and $x_2 \in X$ such that $f(x_1) = f(x_2)$.*

The proof is omitted.†

This material on counting infinite sets may appear very abstract and obscure and indeed it was many years before it gained general acceptance. To give some idea of its power we will conclude by describing one result which Cantor obtained using his ideas.

Definition 14.3.5 *A real number is called* $\boxed{\text{algebraic}}$ *if it satisfies a polynomial equation*

$$x^n + a_1 x^{n-1} + \ldots + a_{n-1}x + a_n = 0$$

with the coefficients a_i all rational. It is called $\boxed{\text{transcendental}}$ *if it is not algebraic.*

Thus all rational numbers are algebraic (since m/n satisfies $nx - m = 0$) and $\sqrt{2}$ is algebraic (since it satisfies $x^2 = 2$). In 1844 the French mathematician Joseph Liouville gave an explicit construction of certain transcendental numbers‡ but it remained a problem to determine whether specific numbers are transcendental. In 1873, Charles Hermite (French) proved that e, the base of the natural logarithms, is transcendental, and these ideas were developed by Ferdinand Lindemann (German) to prove in 1882 that π is transcendental. This result finally proved that the ancient search for a ruler and compass construction to give a square of the same area as a given circle has no solution since it can be shown that any length which can be constructed from a given length is an algebraic multiple of the given length§. These arguments about specific numbers are quite delicate and it is still not known whether, for example, $\pi + e$ is transcendental.

† A proof may be found in D.J. Velleman, *How to prove it, a structured approach,* Cambridge University Press, 1994.

‡ A description of this construction may be found in R. Courant and H. Robbins, *What is mathematics?* Oxford University Press, 1941.

§ See R. Courant and H. Robbins, *op. cit.*

Cantor's remarkable result was that *most* real numbers are transcendental! He proved, using similar methods to the above proof of Theorem 14.2.6, that the set of of algebraic numbers is denumerable (see Problems III, Question 28). It follows from Theorem 14.3.1 and Exercise 14.2 that the set of transcendental numbers is uncountable and it is in that sense that most real numbers are transcendental.

Exercises

14.1 Prove that if A is finite and B is denumerable then $A \cup B$ is denumerable.

14.2 Prove that if A and B are denumerable then $A \cup B$ is denumerable. Hence prove by contradiction that, if X is uncountable and a subset A is denumerable, then $X - A$ is uncountable.

14.3 Prove that if $\{ A_n \mid n \in \mathbb{Z}^+ \}$ is a denumerable set of pairwise disjoint denumerable sets then the union

$$\bigcup_{n \in \mathbb{Z}^+} A_n = \{ x \mid x \in A_n \text{ for some } n \in \mathbb{Z}^+ \}$$

is also denumerable.

14.4 Use the Cantor–Schröder–Bernstein theorem to prove that if $|X| \leqslant |Y|$ and $|Y| \leqslant |X|$ then $|X| = |Y|$.

Problems III: Numbers and counting

1. Of the 182 students who are taking three first year core Mathematics modules (Reasoning, Algebra and Calculus), 129 like Reasoning, 129 like Algebra, 129 like Calculus, 85 like Reasoning and Algebra, 89 like Reasoning and Calculus, 86 like Algebra and Calculus, and 54 like all three modules. How many of the students like none of the core modules?

2. Of the 170 students who took all the first year core modules last year, 124 liked Reasoning, 124 liked Algebra, 124 liked Calculus, 10 liked only Reasoning, nobody liked only Algebra, 4 liked only Calculus and 2 liked none of the modules. How many students liked all three core modules?

3. At an international conference of 100 people, 75 speak English, 60 speak Spanish and 45 speak Swahili (and everyone present speaks at least one of these languages).

(i) What is the maximum possible number of these people who can speak only one language? In this case how many people speak only English, how many speak only Spanish, how many speak only Swahili, and how many speak all three?

(ii) What is the maximum number of people who speak only English? In this case what can be said about the number who speak only Spanish and the number who speak only Swahili?

(iii) Prove that the greater the number of people who speak all three languages, the greater the number of people who speak only one language.

4. Prove (by induction on n) the general inclusion–exclusion principle which may be stated as follows.

Let A_1, A_2, ..., A_n be finite sets. For $I = \{i_1, i_2, \ldots, i_r\} \subseteq \mathbb{N}_n$, write

$$A_I = \bigcap_{i \in I} A_i = A_{i_1} \cap A_{i_2} \cap \ldots \cap A_{i_r}.$$

Then

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\emptyset \neq I \subseteq \mathbb{N}_n} (-1)^{|I|-1} |A_I|,$$

summing over all non-empty subsets of $\mathbb{N}_n$.

5. Prove that a finite non-empty set of real numbers has a minimum element.
[This is the part of the proof of Proposition 11.2.3 left as an exercise.]

6. Suppose that A and B are non-empty finite sets of real numbers such that $A \subseteq B$. Prove that

$$\min B \leqslant \min A \leqslant \max A \leqslant \max B.$$

7. What is wrong with the following argument purporting to prove that for every non-empty finite set A of real numbers $\min A = \max A$?

Proof We prove by induction on n that if $|A| = n$ then $\min A = \max A$.
Base case: If $|A| = 1$ then A is a singleton set $A = \{a_1\}$ and $\min A = a_1 = \max A$.
Inductive step: Suppose that for some positive integer k the result is true for $n = k$. Let A be a set of real numbers with $|A| = k + 1$. Let $a_1 = \min A$ and $a_2 = \max A$. Then

$$\begin{aligned} a_1 &= \min(A - \{a_2\}) \\ &= \max(A - \{a_2\}) \quad \text{(by inductive hypothesis)} \\ &\geqslant \max(A - \{a_1, a_2\}) \quad \text{(by Question 6)} \\ &\geqslant \min(A - \{a_1, a_2\}) \quad \text{(by Question 6)} \\ &\geqslant \min(A - \{a_1\}) \quad \text{(by Question 6),} \\ &= \max(A - \{a_1\}) \quad \text{(by inductive hypothesis)} \\ &= a_2. \end{aligned}$$

Hence $a_1 \geqslant a_2$ and since clearly $a_1 \leqslant a_2$ this proves that $a_1 = a_2$ as required to deduce the result for $n = k + 1$ and so proving the inductive step.
Conclusion: Hence by induction on n, $\min A = \max A$ for a set of cardinality n for all positive integers n, and so for all finite sets. □

8. Prove Lemma 10.1.4 by induction on m.

9. Prove Proposition 11.1.4: if $X \to \mathbb{N}_n$ is an injection, then X is finite and $|X| \leqslant n$.

10. Prove Theorem 11.1.6: given non-empty finite sets X and Y with $|X| < |Y|$ there does not exist a surjection $X \to Y$.

11. Use the pigeonhole principle and proof by contradiction to prove Theorem 11.1.7: given non-empty finite sets X and Y with $|X| = |Y|$, a function $X \to Y$ is an injection if and only if it is a surjection.

12. Suppose that there is an injection $f: \mathbb{Z}^+ \to X$. Prove by contradiction that X is an infinite set.

[Use Corollary 11.1.1 noting that, for any $n \geqslant 1$, f restricts to give an injection $\mathbb{N}_{n+1} \to X$.]

13. Find all the divisors of 126 and 180 and hence find the greatest common divisor $(126, 180)$.

14. For $n \in \mathbb{Z}^+$, suppose that $A \subseteq N_{2n}$ and $|A| = n + 1$. Prove that A contains a pair of distinct integers a, b such that a divides b.

[Let $f(a)$ be the greatest odd integer which divides a and apply the pigeonhole principle to f.]

15. Prove the induction principle from the well-ordering principle (see Example 11.2.2(c)).

[Prove the induction principle in the form of Axiom 7.5.1 by contradiction.]

16. Use the inclusion–exclusion principle (Question 4) to prove that the number of surjections $\mathbb{N}_m \to \mathbb{N}_n$ is given by

$$n^m - \binom{n}{1}(n-1)^m + \binom{n}{2}(n-2)^m - \ldots + (-1)^{n-1}\binom{n}{n-1}1^m.$$

[For each $i \in \mathbb{N}_n$, let $A_i = \{ f: N_m \to \mathbb{N}_n \mid i \text{ is not a value of } f \}$.]

Deduce that

$$n^n - \binom{n}{1}(n-1)^n + \binom{n}{2}(n-2)^n - \ldots + (-1)^{n-1}\binom{n}{n-1}1^n = n!.$$

17. A *derangement* of $\mathbb{N}_n$ is a bijection $f: \mathbb{N}_n \to \mathbb{N}_n$ with no fixed points, i.e. $f(i) \neq i$ for all $i \in \mathbb{N}_n$. Use the inclusion–exclusion principle to prove that the number of derangements of $\mathbb{N}_n$ is

$$n!\left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \ldots + (-1)^n \frac{1}{n!}\right).$$

[This means that the proportion of all permutations of $\mathbb{N}_n$ which are derangements is $1/2! - 1/3! + 1/4! - \ldots + (-1)^n 1/n! = 1 - 1/1! + 1/2! - 1/3! + 1/4! - \ldots + (-1)^n 1/n!$. The reader may recognize this as the first part of the series for e^{-1}. It follows that for large values of n the probability that a permutation is a derangement is approximately $1/e$.]

18. Suppose that X and Y are disjoint sets. Prove that the function

$$\bigcup_{i=0}^{k} \mathcal{P}_i(X) \times \mathcal{P}_{k-i}(Y) \to \mathcal{P}_k(X \cup Y)$$

defined by $(A, B) \mapsto A \cup B$ is a bijection. Deduce that

$$\binom{m+n}{k} = \sum_{i=0}^{n} \binom{m}{i}\binom{n}{k-i}.$$

Give an alternative proof of this result by induction on n using Proposition 12.2.8.

19. Prove Leibniz's rule for higher order derivatives of products,

$$\frac{d^n(uv)}{dx^n} = \sum_{i=0}^{n} \binom{n}{i} \frac{d^i u}{dx^i} \frac{d^{n-i} v}{dx^{n-i}} \quad \text{for } n \in \mathbb{Z}^+,$$

by induction on n.

20. Use the pigeonhole principle to prove that, given ten distinct positive integers less than 107, there exist two disjoint subsets with the same sum.

21. Define the notion of inequality between two fractions by

$$\frac{a_1}{b_1} < \frac{a_2}{b_2} \Leftrightarrow \begin{cases} a_1 b_2 < a_2 b_1 & \text{if } b_1 b_2 > 0, \\ a_1 b_2 > a_2 b_1 & \text{if } b_1 b_2 < 0. \end{cases}$$

Prove that this definition leads to a well-defined notion of inequality between two rational numbers satisfying Axioms 3.1.2.

22. For a positive real number x and a rational number $q = m/n$ (where m and n are integers), define the power x^q to be $(x^{1/n})^m$. In this expression the integral power is defined in Problems I, Question 23 and $x^{1/n}$ is defined in Example 9.2.4(b).

Prove that this definition of rational powers is well-defined. With this definition prove the laws of exponents for any positive real numbers x and y and rational numbers p and q:

(i) $x^q y^q = (xy)^q$;
(ii) $x^{p+q} = x^p x^q$;
(iii) $(x^p)^q = x^{pq}$.

23. Prove that there does not exist a rational number whose square is 10.

24. Find the first four decimal places of the infinite decimal representing $\sqrt{10}$ (without using a calculator!).

25. Find the rational number equal to the recurring infinite decimal $2.1001\dot{2}09\dot{7}$.

26. Prove that each finite decimal may be written as an infinite decimal in two distinct ways:

$$\begin{aligned} a_0.a_1a_2\ldots a_{n-1}a_n &= a_0.a_1a_2\ldots a_{n-1}a_n\dot{0} \\ &= a_0.a_1a_2\ldots a_{n-1}(a_n-1)\dot{9}, \end{aligned}$$

where $a_n > 0$ if $n > 0$.

27. Prove that each real number is represented by a unique infinite decimal unless it is representable by a finite decimal, in which case it is representable by precisely two infinite decimals as described in the previous problem.

[Hint: Suppose that two infinite decimals represent the same number: $a_0.a_1a_2\ldots a_n\ldots = b_0.b_1b_2\ldots b_n\ldots$. Attempt to prove by induction that $a_n = b_n$ for all $n \geqslant 0$.]

28. Prove that the number of polynomials of degree n with rational coefficients is denumerable. Deduce that the set of algebraic numbers (see Definition 14.3.5) is denumerable.

[Recall that a polynomial equation $x^n + a_1x^{n-1} + \ldots + a_{n-1}x + a_n = 0$ has at most n real roots.]